Back to Basics - Technical Report

A technical report examining site climate, hempcrete material performance and detailed construction for Back to Basics, an off-grid hemp housing scheme in Hackney Wick Fish Island. Macro and micro climate analysis, case studies and daylight and thermal testing inform a hempcrete wall detail that performs passively across the seasons.



01/01/2023 - date started

15/03/2023 - date ended

Macro and micro climate analysis, precedent case studies and daylight, thermal and shadow testing assess whether hempcrete can perform as both a material and environmental system, concluding in a detailed wall section and passive, off-grid design strategy.

Documents by Sebastian Dawber
https://www.sebastianvictordawber.co

Created for ...
eg BSc Architecuture and enviromental design
Undergraduate RIBA ARB part I

All work produced by Sebastian Victor Dawber

https://www.sebastianvictordawber.co/

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Https://www.instagram.com/sarc_studio/



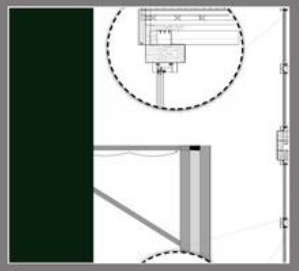
SEBASTIAN VICTOR DAWBER
Independent Project
BSc Part 1, ARB RIBA , Architecture and environmental design

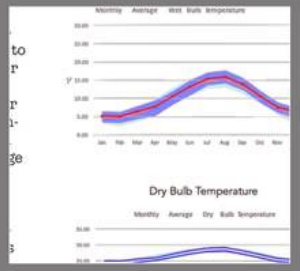
Back to Basics - Technical Report

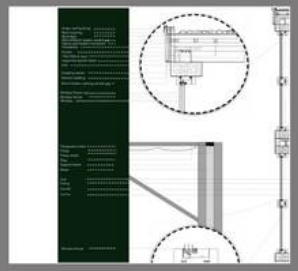
Climate Analysis, Hempcrete Performance & Detailing

15/03/2023 - date ended

This document is the technical report supporting Back to Basics, an off-grid hemp housing scheme for Hackney Wick, Fish Island. It sets out a methodology for understanding the site through macro, micro and specific-site climate analysis, testing environmental conditions across the year to identify site constraints and opportunities. Case studies, including a primary healthcare centre near Dharmapuri, Ameri House in Kashan, a Mulberry/SHoP Architects scheme and the Canabric single-family house, are used to test whether hempcrete can perform as a material system. Shadow, daylight and thermal strategies are developed and tested against the site's climate, culminating in a detailed hempcrete wall section. The report concludes that hempcrete's hydroscopic nature allows a closed-plan brick wall to regulate air and temperature through the seasons, supporting a passive, off-grid building strategy for Hackney Wick.



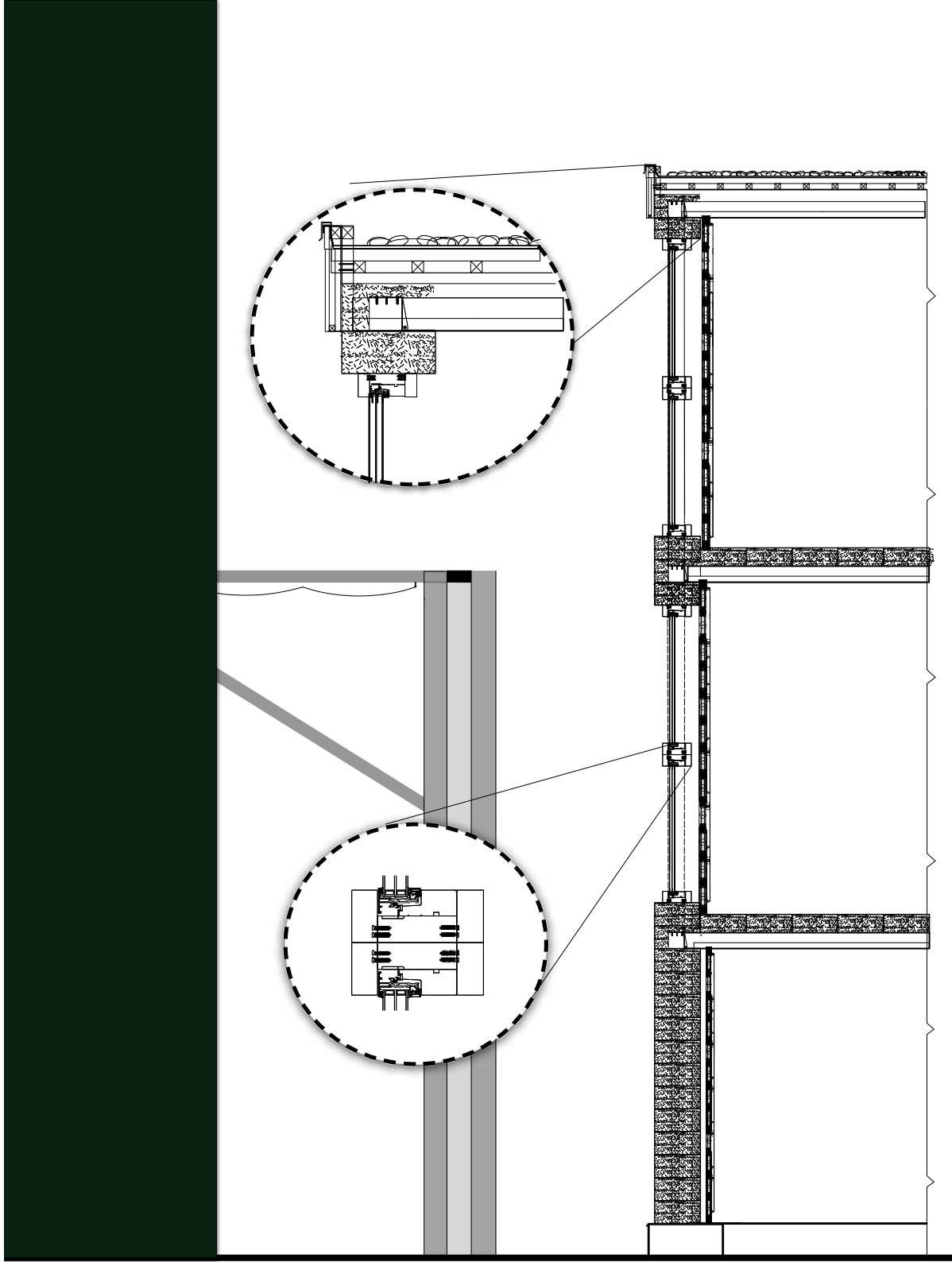




Techncial report

Sebastian Victor Dawber

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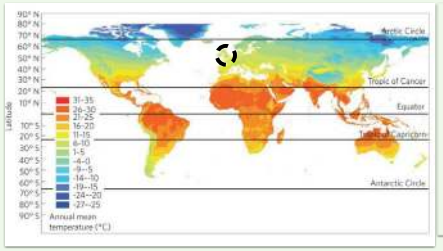
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Objecttives

- To conduct a site macro, micro and specific site analysis to understand the entire area
-
- Using a methodology to understand Hackney Wick Fish Island
-
- Simulate various environmental conditions throughout the year
-
- Schematic drawings to show how it works and where the methodology is taking place
-
- Analyse the proposed building material to see its performance and weather of not it can be used as a material system.
-
- Asses whether hemp can be used a material system and if the strategies found in the precedents can work effectively.

1









Online research was done to develop an understanding of London's climate and the site. Precedents were found during which the start of my design was formed. The research links and brings my portfolio into one whole body of work.

2



Anemometer



Acoustic meter



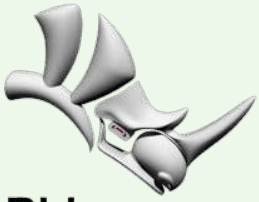
Laser meter



Compass

Field testing was conducted to gather first hand research to understand the micro climate of the site

3





Rhinoceros 3D was used to create digital models to be used for an environmental analysis to assess the performance of hempcrete as well as the design strategies. Once created IES VE was used to conduct a environmental performance analysis.

4



Graphic

The final presentation was made using graphic leading to the end of the report

Introduction

Bioclimatic research to provide problems, nature to provide solutions. The intention is to show proof that hemp can be used as a material system. Various intentional were found upon research, strategies to create an off-grid natural green building. Using passive and active strategies to reduce nervy demand as well as collect natural resources.

The methodology provided various results all leading to one solution, using evidence-based testing one can find results to inform design decisions. Integrated environmental solutions were developed from software to provide accurate results, with added first-hand site research. A green issue solved with green solutions. Flowing resources into a collection, a circular economy where people meet people, and people meet nature.

Concluding to assess the effectiveness of hempcrete and design strategies found based on research and methodology. A plant from the past using ways of the past, to change the past to come.

1.Climate analysis
London
51.5074° N, 0.1278° W

Elevation: 35 m
Ecozone: Palearctic
Climate: Oceanic

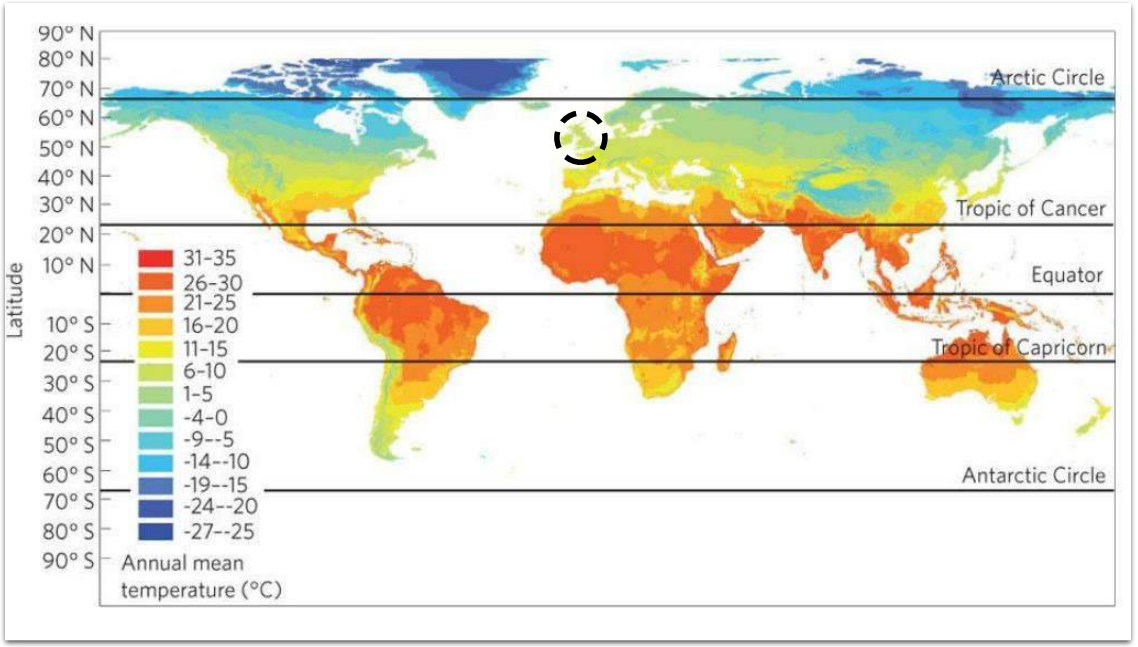
Seasons:
Winter(December - February) 22 December winter solstice: Temperatures: 6.6(C)- 7.4(C) Daylight Hours: 8-9 hours Monthly Rainfall (mm): 78 mm
Spring(March-May)

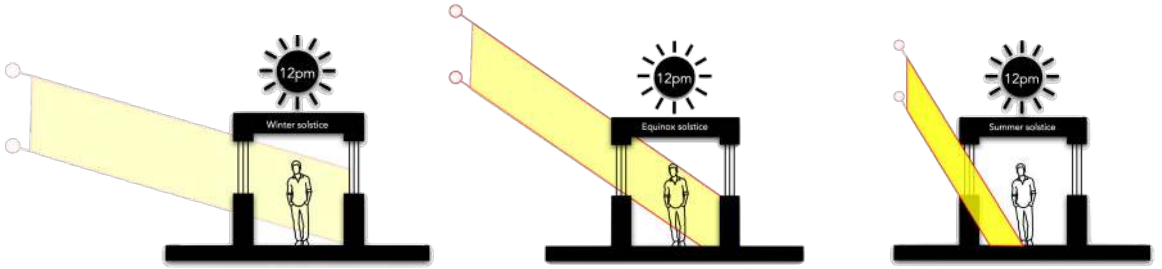
22 March equinox solstice: Temperatures: 9.3oC- 15.4(C) Daylight Hours: 11-15 hours Monthly Rainfall (mm): 60 mm

Summer (June to August)
22 June summer solstice: Temperatures: 18.1(C)- 20.6(C) Daylight Hours: 16 hours - Rainfall (mm): 61 mm

Seasonal changes lead to changing environmental conditions. Low sun angles in the colder months with short days and stronger winds. Higher sun angles in summer with longer days and cooling winds. The main issue for the United Kingdom is lack of sun and cold temperatures in the winter season, bright sun angles and high temperatures in the summer season. Using solar and thermal control, the negative adverse effects of climatic conditions will be reduced.

United Kingdom, London.

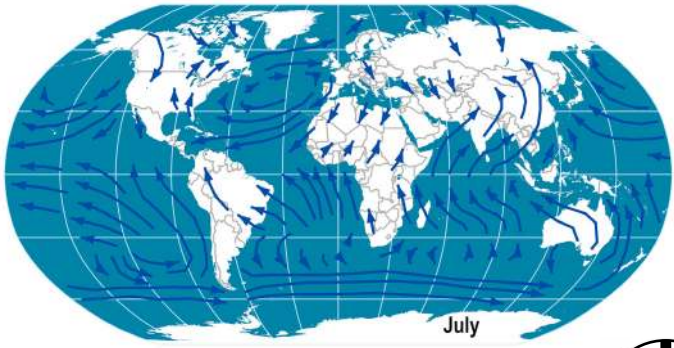






South West wind direction



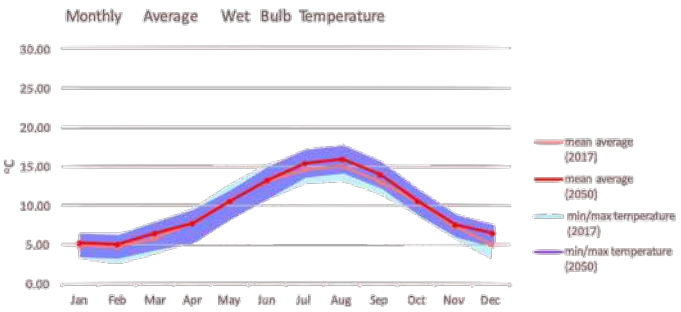


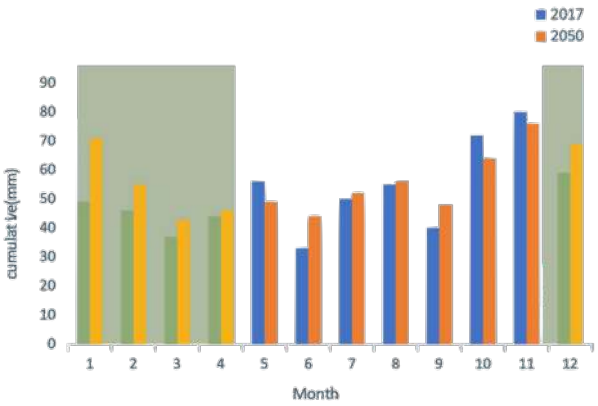
North East wind direction

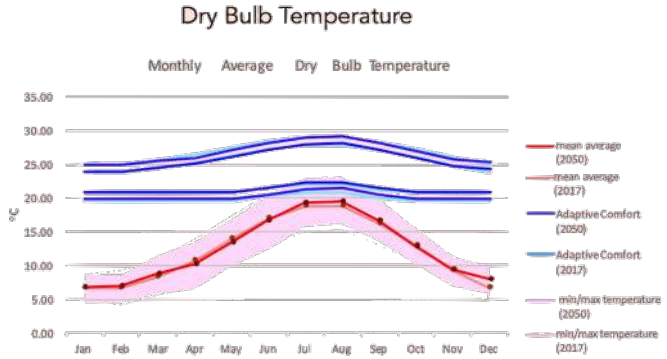


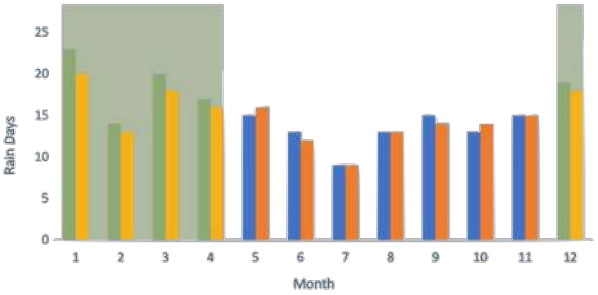
London, as a city, faces many environmental challenges. Due to its longitude and latitude, seasonal changes lead to short sun hours in the colder months. The sun angle also changes throughout the year with lower sun angles in winter and higher sun angles in summer. Temperatures range from 6°c in winter to 21 °c in summer, which is affected by annual wind changes throughout the year. In winter the South Western winds bring cold drafts through the city and in summer the North Eastern winds cool down the city.

The main issue for the United Kingdom is the lack of sun and cold temperatures. Through passive solar and thermal design, these issues will hopefully be reduced.

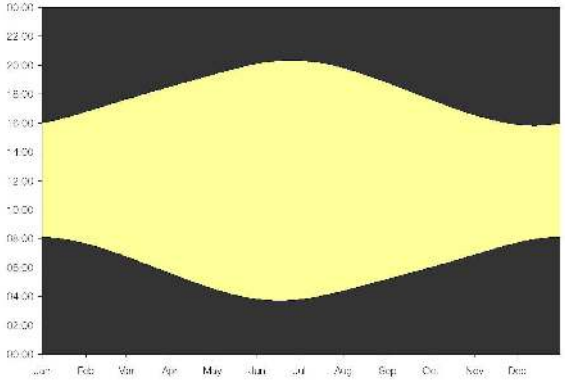




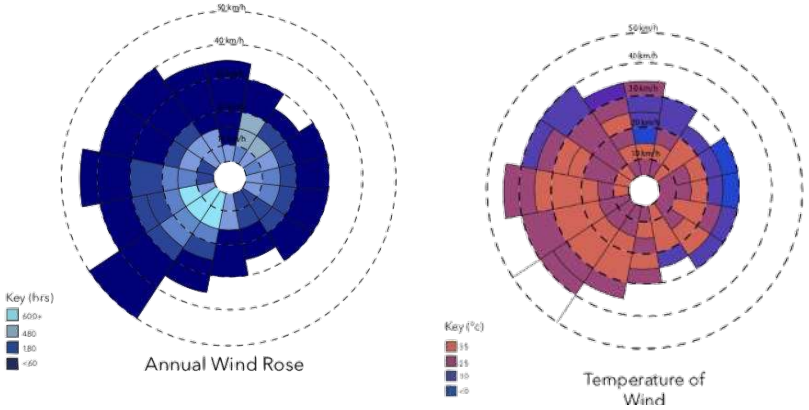




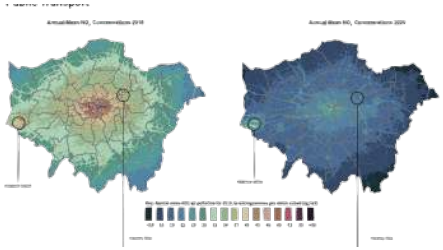
Annual sun hours (United Kingdom)



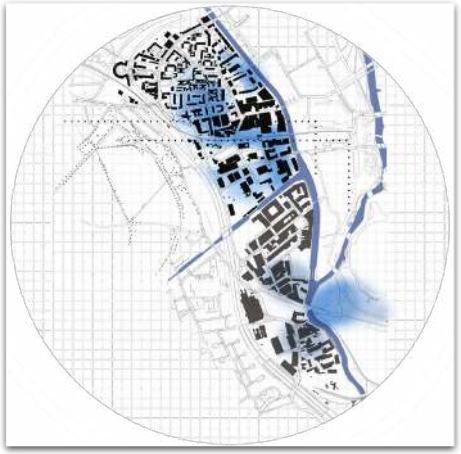
Annual Wind direction (United Kingdom)



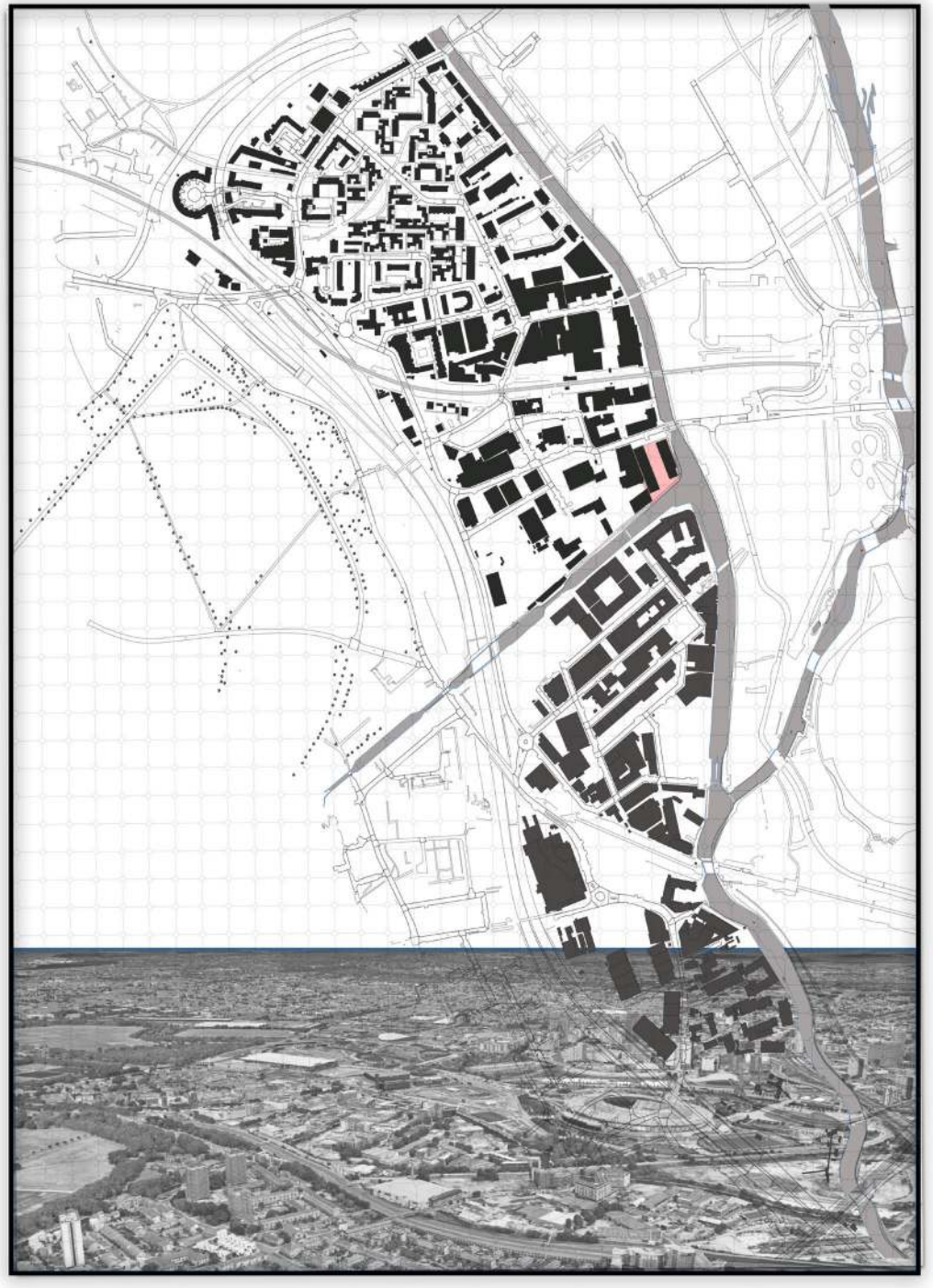
The overall climate presents a number of challenges. Looking specifically at the site as a whole, a macro climate analysis was conducted to asses issues effecting the intended design. Concerns arose and led to design strategies that followed the overall climate issues. The start of the environmental site analysis.

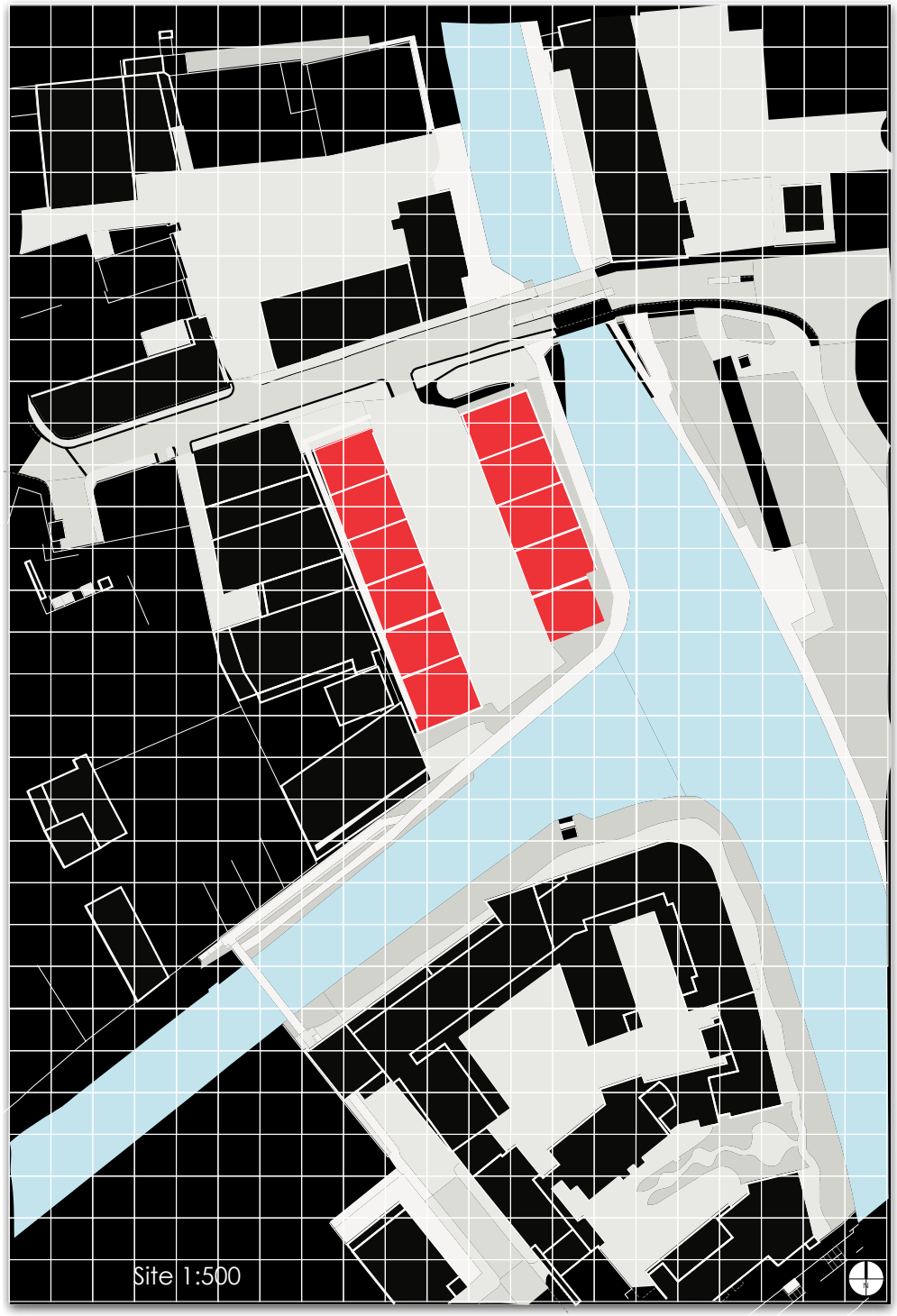


Major issues affecting the site such as pollution levels, flooding and no central green spaces led the start of the design ideas. The issues branch onto one another and possible sorties to deal with all are possible. A green space the absorbs surrounding pollution levels while collecting and redistributing water may be an option—an area centrally located in the heart of Hackney Wick Fish Island to solve issues listed. The diagrams show where and how these effects came into play and were the start of the site mapping to place the design in the best position while adhering to the earthy of a green space that redistributes water while absorbing large amounts of pollution.



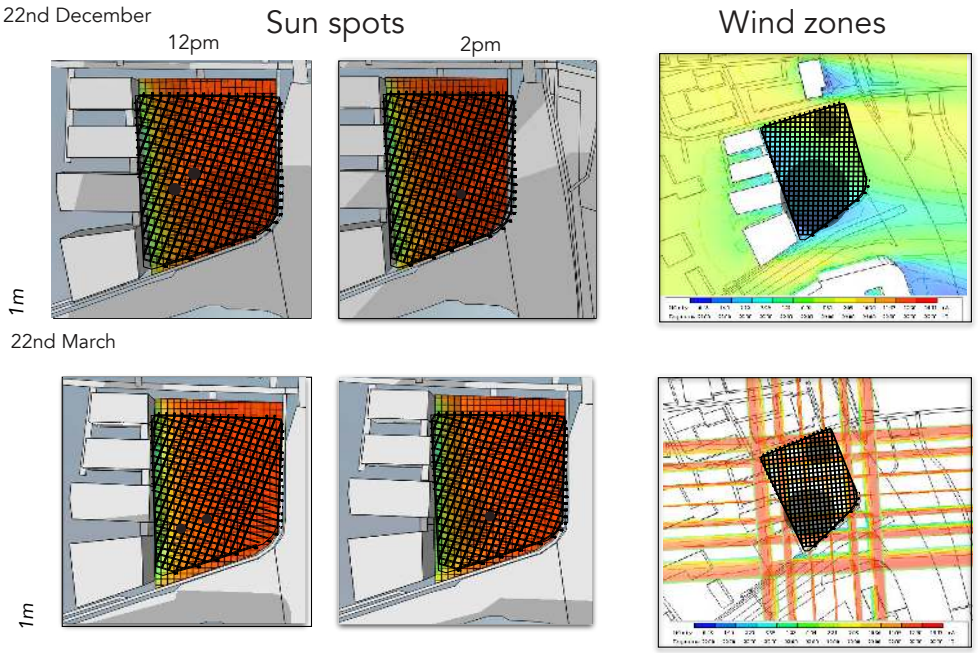






The macro analysis gave way to the start of possible site strategies. Once the macro issues were analysed, more in-depth research into the site was conducted. A micro site analysis to pose further constraints and opportunities within the chosen location.

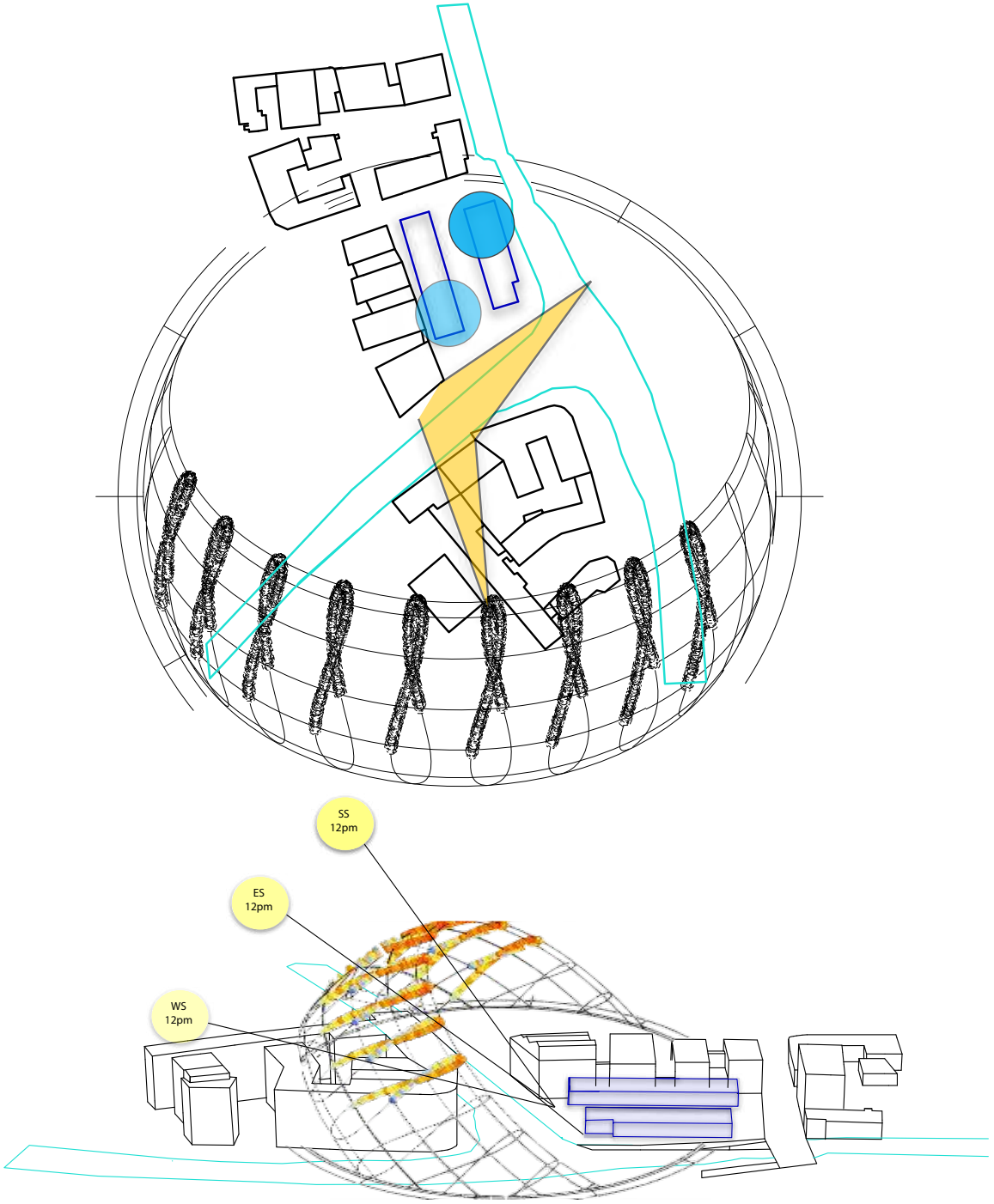
Main issues analysed gave ideas to form the design. Thermal conditions led to fluctuating wind directions. The climate analysis of London, showed seasonal changes in wind direction, keeping this in mind; the surrounding context created further changes in wind directions. A zone where winds speeds are reduced was found. Possible placement in this zone would reduce the impact when wind speeds are high but will still allow for passive strategies to occur; thus, the wind effects would lead to the placement of the form. Solar issues due to overshadowing caused by neighbouring context meant light is blocked during certain times and seasons. Vital for the growing of crops and passive solar design, testing began at different levels and seasons to find sunspots and find where the functions should be placed that would allow for the sun to enter naturally lighting the space. Various level testings revealed shadow zones, if they were constant from 6m-3m then any design placed in this zone would have no overshadowing on the external and surrounding spaces, the placement of functions was thus developed from the solar testing.



To sum up

Wind speeds fluctuate, but zones with lower rates were discovered, the state of the form placement.

The analysis showed deflected light and over-shadowing, as well as sunspots that arise due to the surrounding context. Sunspots to allow the sun to enter the central courtyard between 12-2 pm were the result of a shadow and radiation analysis, as mentioned previously. Due to site activities taking place between 12-2 pm.It also allowed context shadows to be exposed and any design placed in shadows between 1 to 3 m means the implemented design won't cause overshadowing due to it being placed in existing shadows.







The site itself presents both constraints and opportunities. The constraints are as follows: Surrounding buildings blocking out the sun, wind, downdraft and wake effects, high humidity levels near the water, high pollution levels near the road and fluctuating wind effects due to the surrounding high rise buildings.

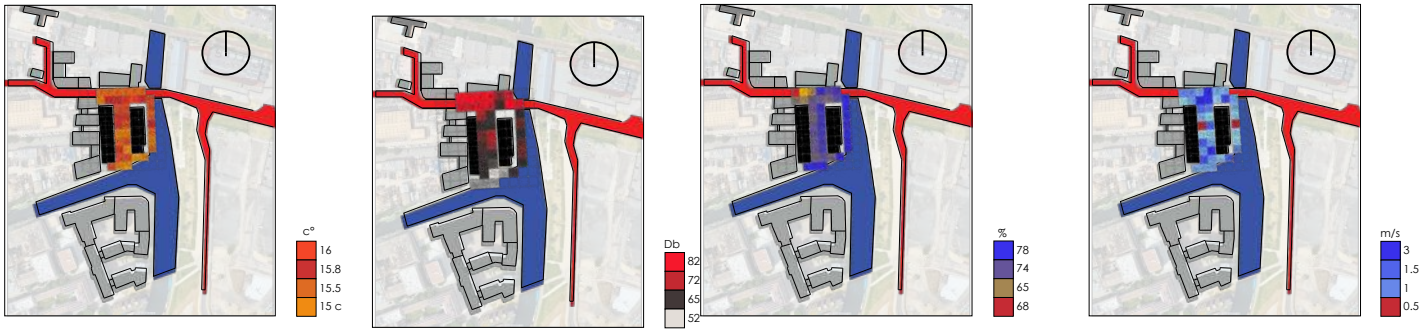


-South and West side adjacent buildigns block direct sunlight on the site

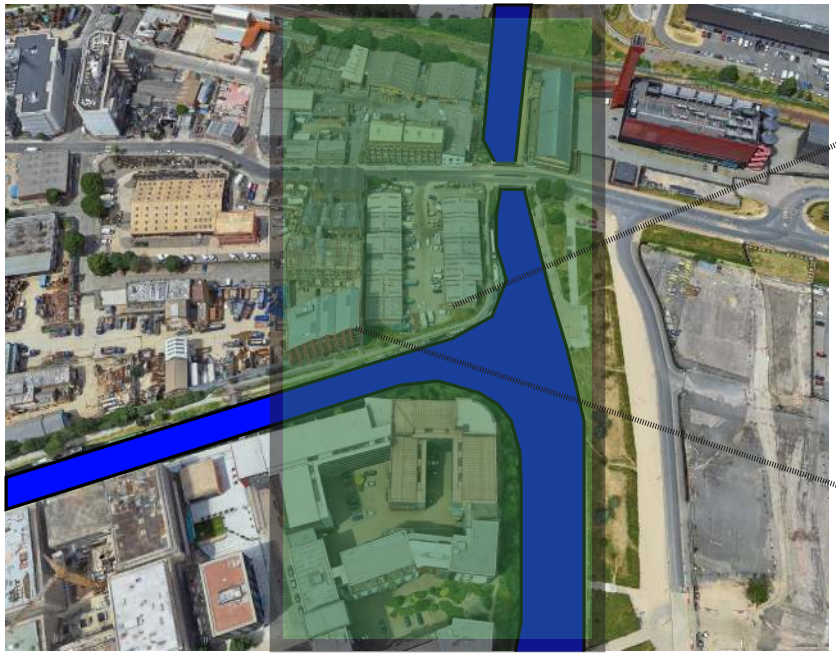


-Wind tunnels are created down the centre of the site

Low sun angles in winter lead to less sun entering the site. Fluctuating wind patterns drive wind down the site creating a wind tunnel due to the placement of the existing buildings

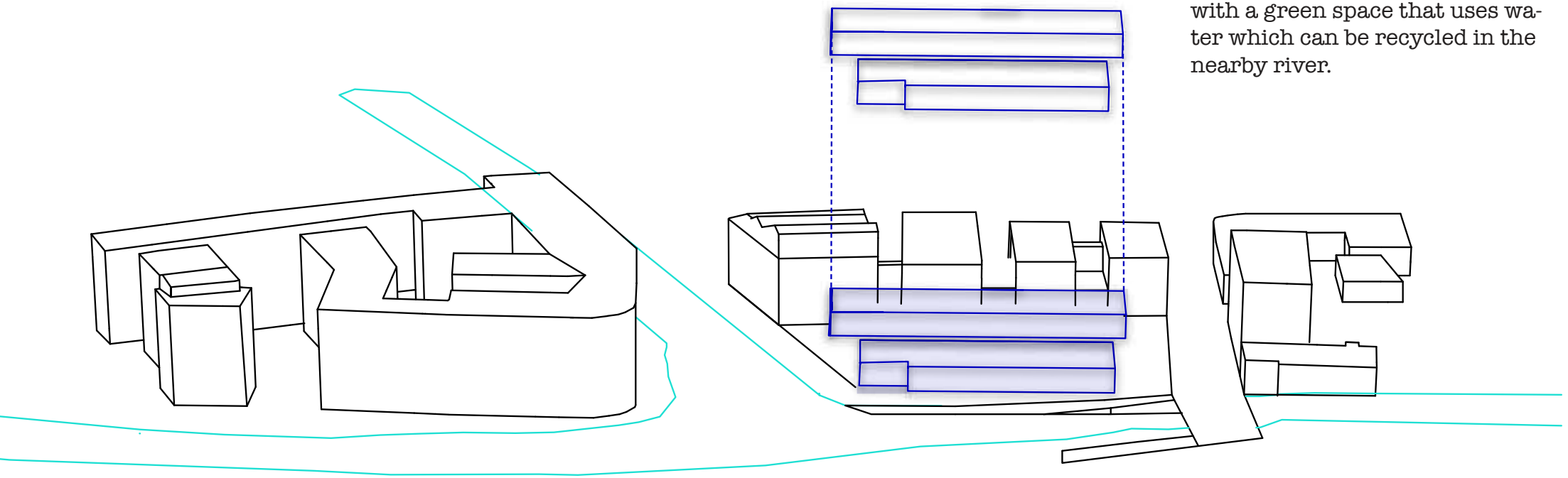


Temperature is cooler near the river, pollution is higher near the road, humidity is high near the river and wind speeds fluctuate due to the site surroundings and existing buildings.



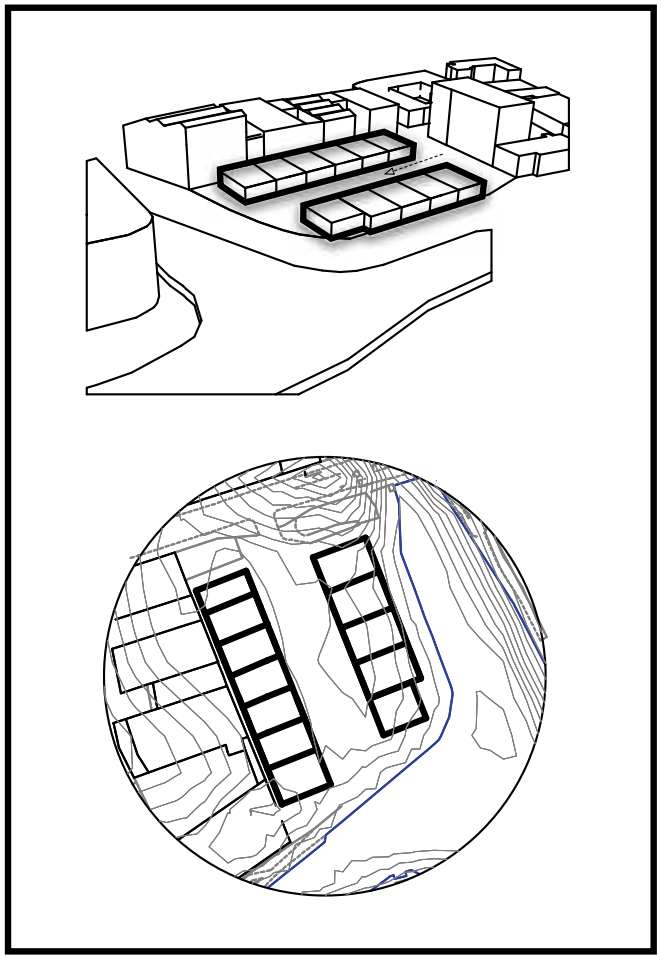


The opportunities of the site gave me the starting point of my design. Located near a river with green areas surrounding it and open views of the Queen Elizabeth Olympic Park. The 5400 sqm site is used as an industrial site; hence its name Hamlet industrial centre. A site with a large space located in a central position near both Hackney Wick and Fish Island. To serve as a bridge between the two places bringing people together through the form and function of the design, starting with a green space that uses water which can be recycled in the nearby river.

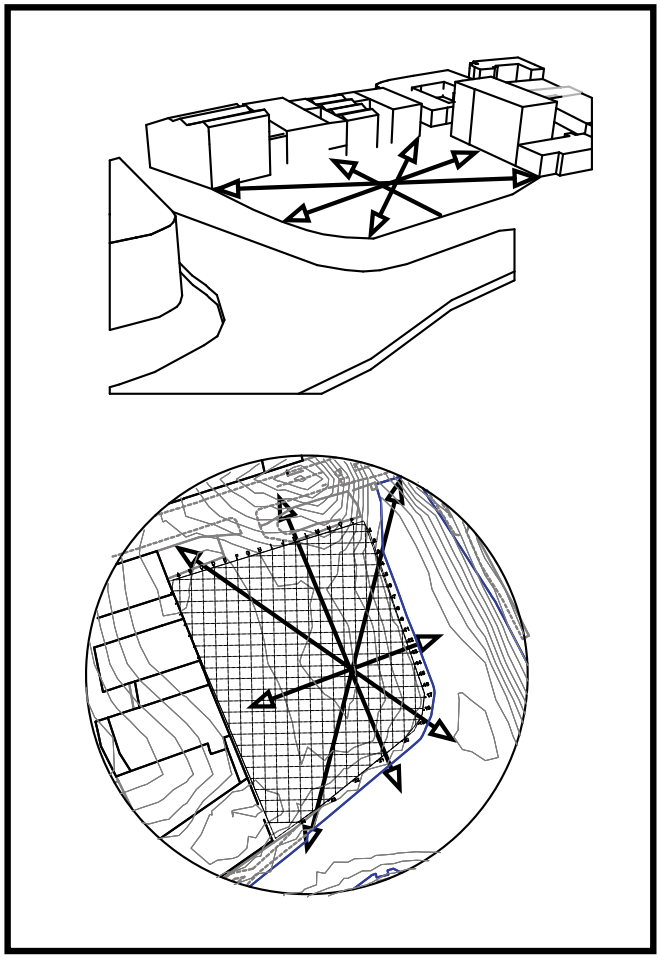


The site has several different activities. A cafe, fresh food store, architectural services, carpentry services and a furniture store. All spaces will be kept but redistributed on site to save space and prevent space waste. Plans to demolish the site make it perfect for any design to be implemented with ease. Allowing for new entrances and connection points to be added. Once demolished a grid was developed to realign the site, so it is exactly south facing. Using a sun path diagram lines were shirt ?the site so it runs exactly from North to South on a linear line. This will allow for passive solar and thermal controls to be implanted with accuracy as windows and strategies can be placed on the North-South axis or East-West axis maximising passive design strategies. It will be used to distribute spaces and organise the site as well.

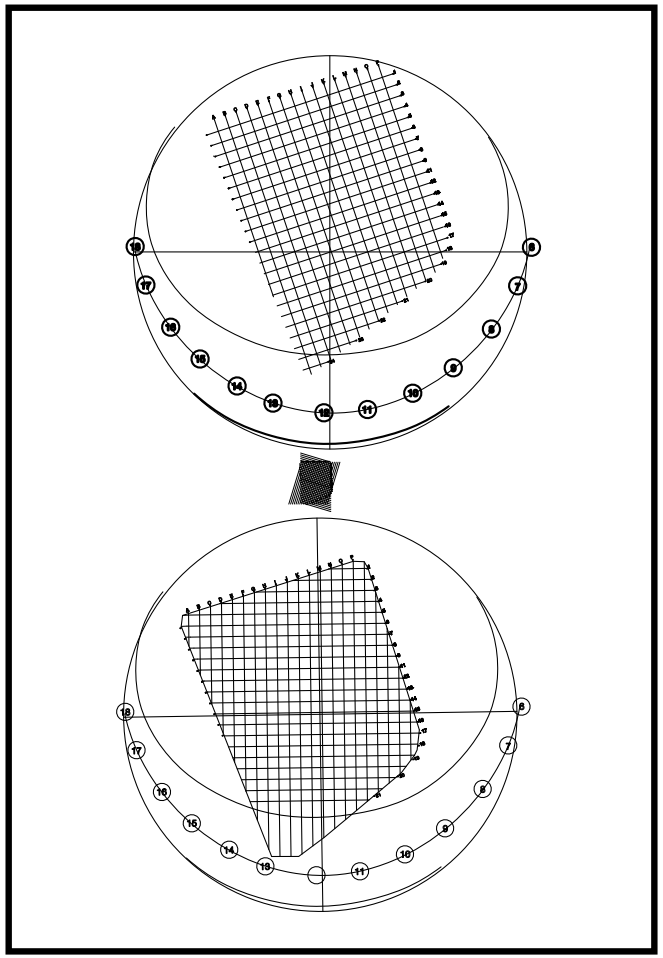
1.



2.



3.



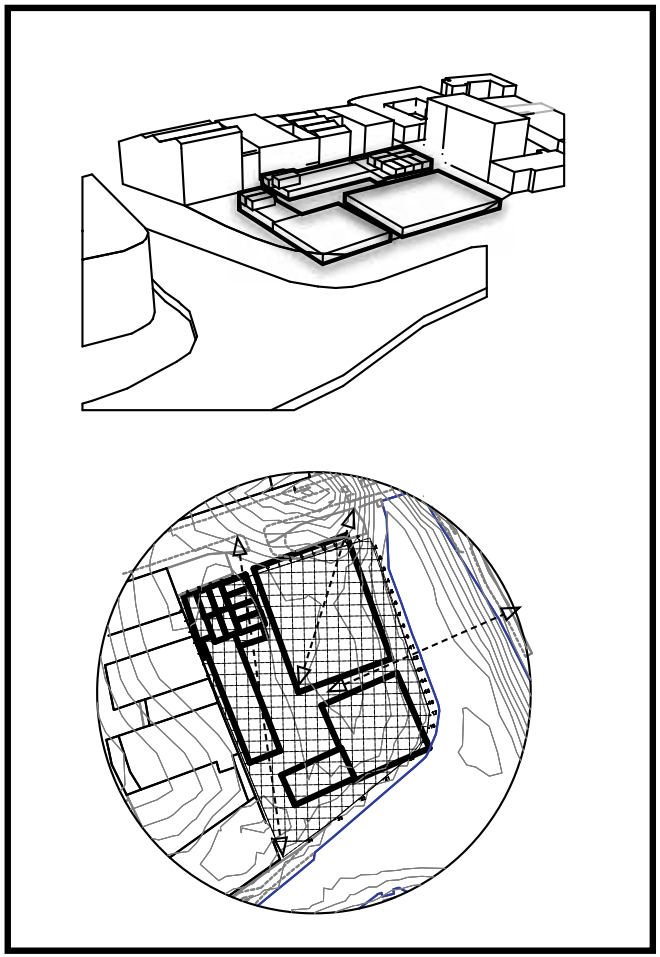
1.
Exisiting site.
- 8 spaces at 176sqm per space.
-Total site space 5800 sqm
-One entracnce

2.
Existing site demolished
-Total site area space 5800 sqm
-Multiple entrances added to create as much access possible
-Connect the two sides of hackney wick fish island

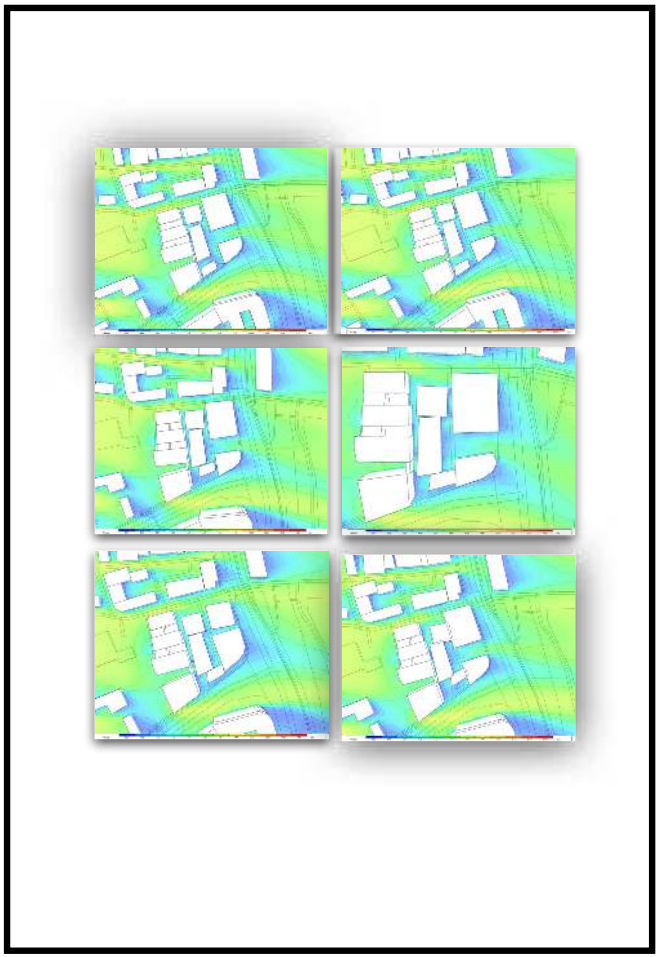
3.
Existing site demolished
Grid used to realign the site using a sun-path digram to rotate the site so the site runs align North to South.

Back to basics seeks to bring people together and create a green productive space where people can meet and learn about the process of making hemp. The spaces offered will be: cafe, growing spaces, work spaces, processing spaces, learning spaces and living spaces. Differing in size, volumes were made to allow for a volumetric form find process that will eventually turn into actual spaces. Wind testing started off the process. Placing the volumes to find the best placement that will allow for passive thermal design and to allow wind to flow freely without creating turbulent zones. The Starting iteration turned out to be the best with elements from the others a wind volumetric form started to develop. The spaces between the volumes allow for wind to pass through the site with ease. The placement for the volumes was found and access points were added.

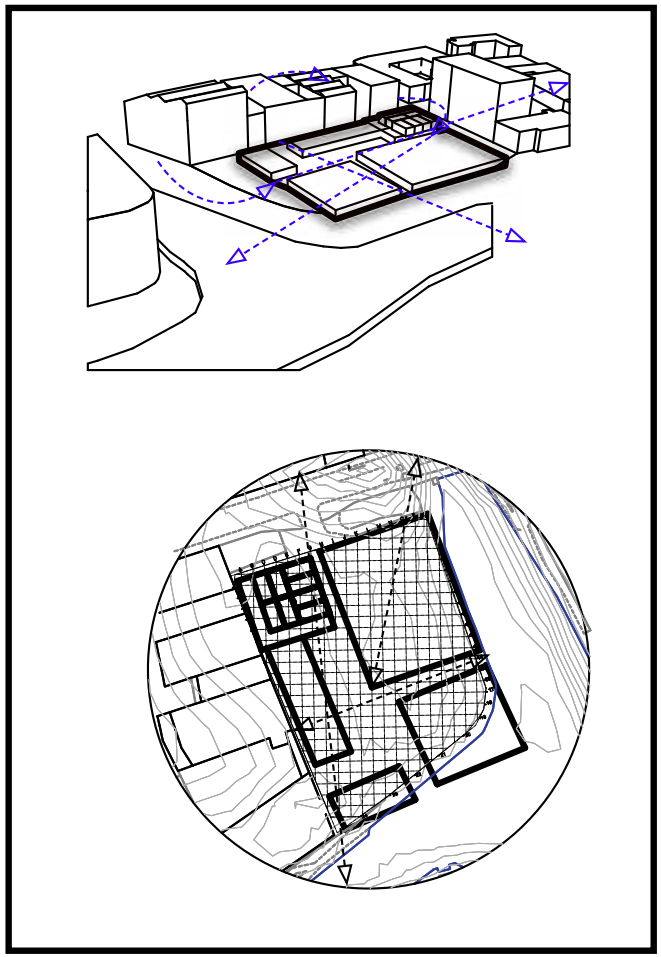
4.



5.



6.



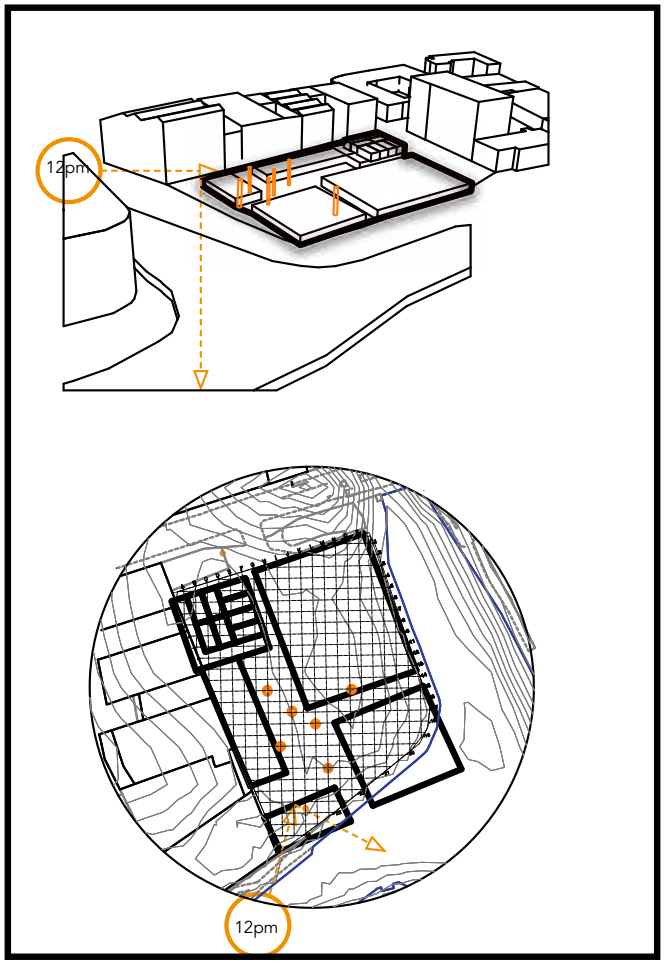
4.
Exisiting site.
- 8 spaces at 176sqm per space.
-Total site space 5800 sqm
-One entracnce

5.
Wind testing using CFD on IES VE. Starting in their original positions the volumes were tested to see how wind reacts with the site and proposed volumetric spaces

6.
Gaps between the volumes allow wind currents to travel freely without creating turbulent zones

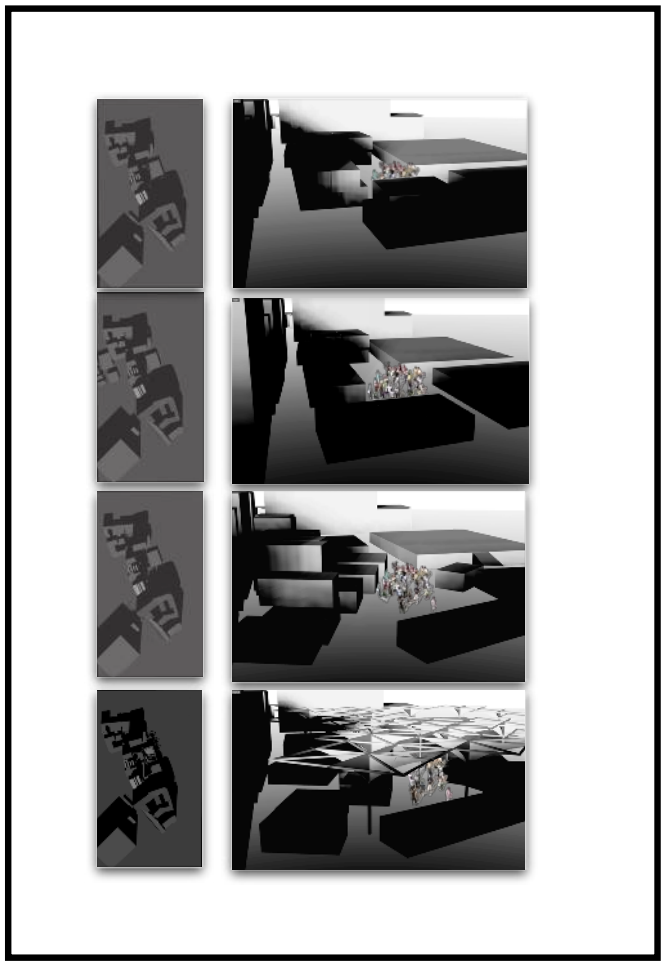
Once the wind volume was formed sunspots were added to allow light to enter the site at specific seasons throughout the year. Learning activities in the café happen at 12 pm thus light is needed in the centralised courtyard. Shadow testing for all the volumetric iterations shows light entering the site at 12 pm in all seasons. Confirming the sunspots allowed for a sun volumetric form to be made. Once moved to fit around the sunspots the sun volumetric form was created, the learning space was joined with the café and a gap between the learning space and processing space allowed for light to pass through. People will be able to enjoy direct light while partaking in hemp processing activities. The volumetric form was created.

7.



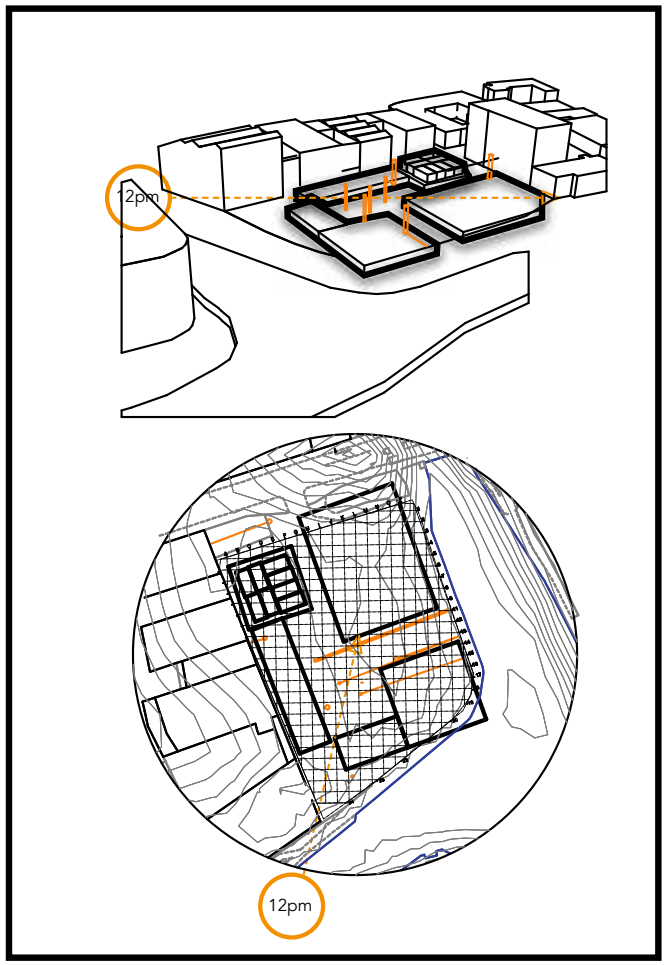
7.
Sun spots were added to allow light to enter at 12m in all seasons

8.

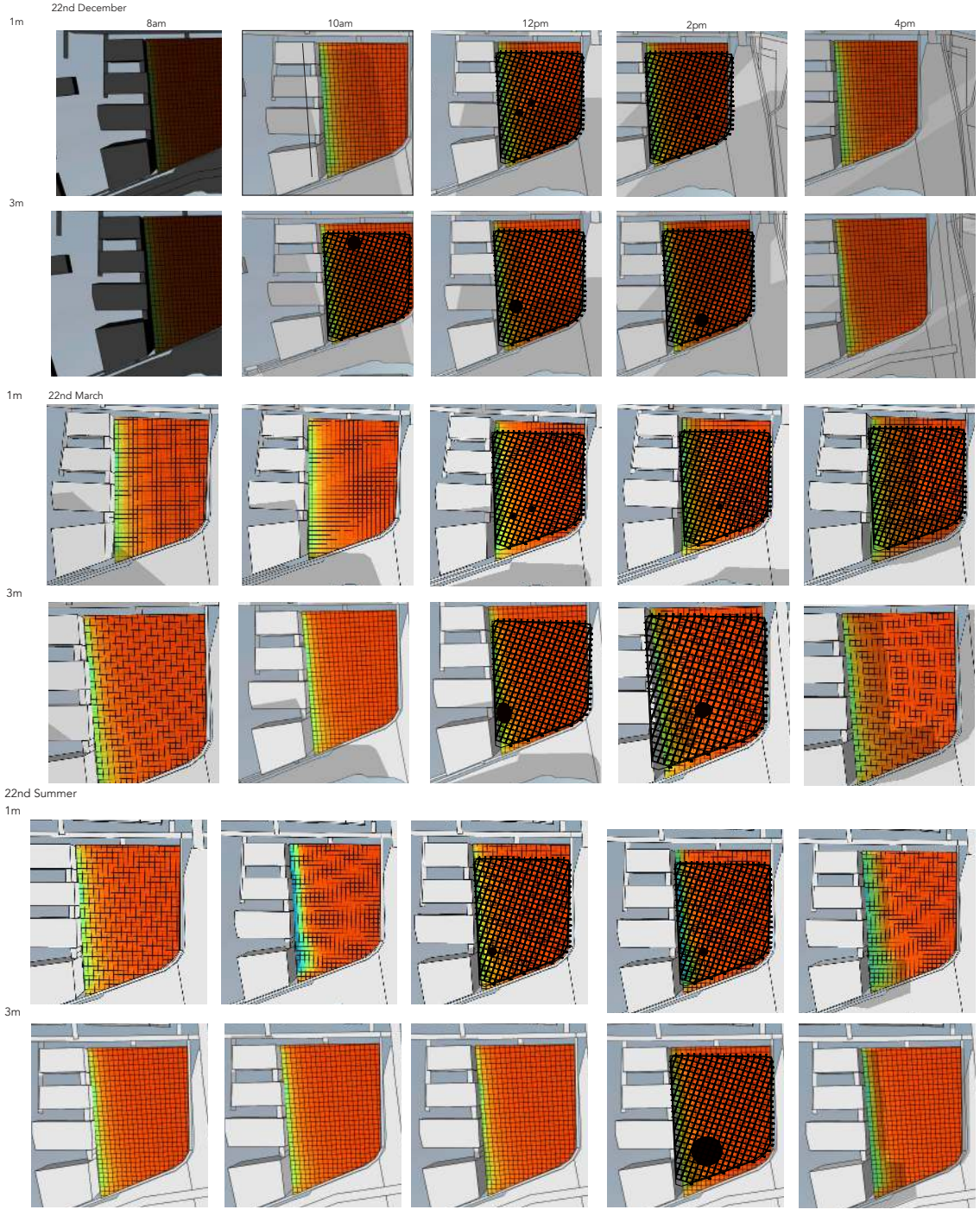


8.
Shadow testing to see the accuracy of the sun spots. Allowing light to enter the central court yard at 12pm

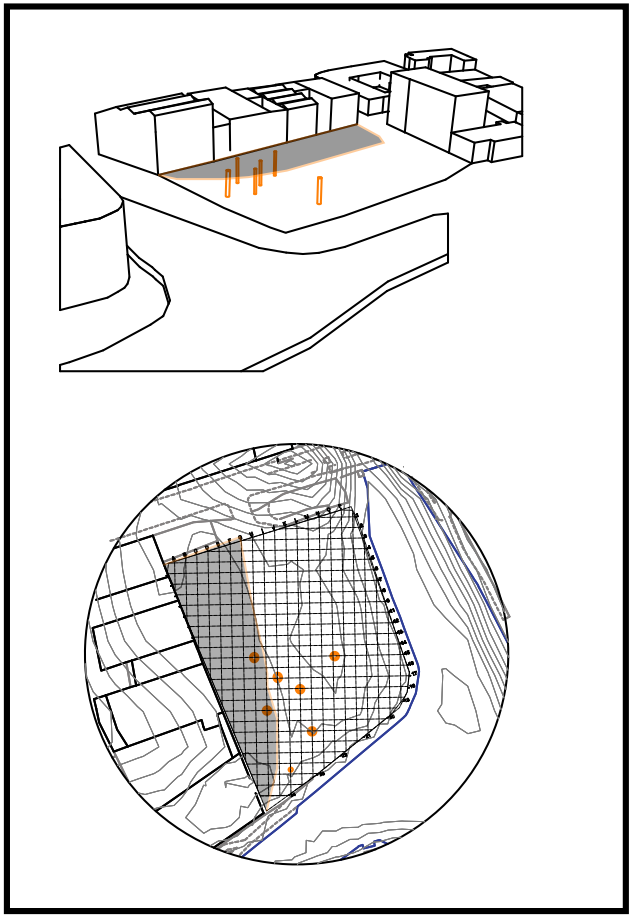
9.



9..
Direct sun light enters the site at 12pm and 2pm in all seasons



IESVE was used to generate a series of analysis, looking a shadows and solar radiation. Using 1m and 3m heights the analysis allowed me to see the site at different levels and where the shadows are. If the shadow is the same at 3m and 1m then any building placed in that area will have little or no overshadowing at the given time, this was the diver for the analysis and allowed me to place building in areas that would not over shadows others; working within the shadows to find sun spots and shadow zones









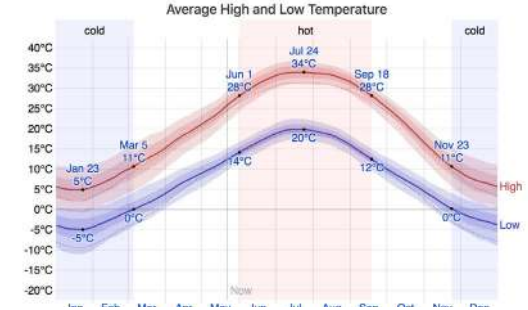


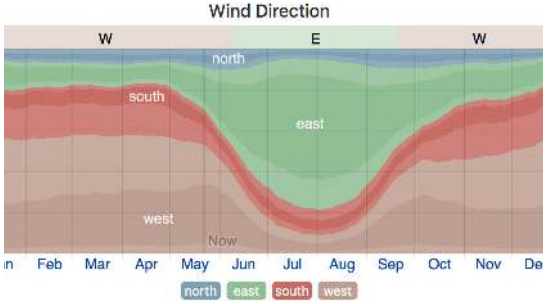
https://www.lafargeholcim-foundation.org/media/news/projects/primary-healthcare-center-near-dharmapuri-india

Primary healthcare centre, near Dharmapuri, India

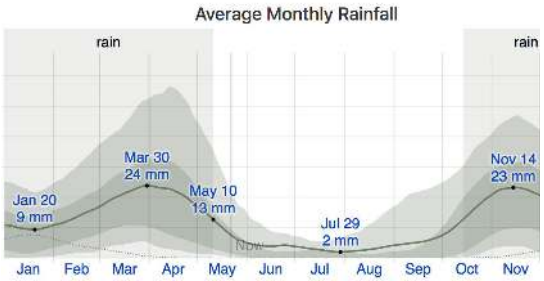
Using Green design features, the health centre located in the rural parts of India uses low tech technology to reduce energy demand. Thatch panels are placed on the side woven with grass to cool the building off, and due to their louvre system, they can change according to weather patterns.

The main feature that I incorporated was the roof, due to high rain in the UK's winter season the idea is to mimic the roof system used in this design which channels rainwater into a concrete gutter well for water collection. The feature will allow my design to use as little water as possible and collect as much free water as possible. This system was applied to the first-floor roof.

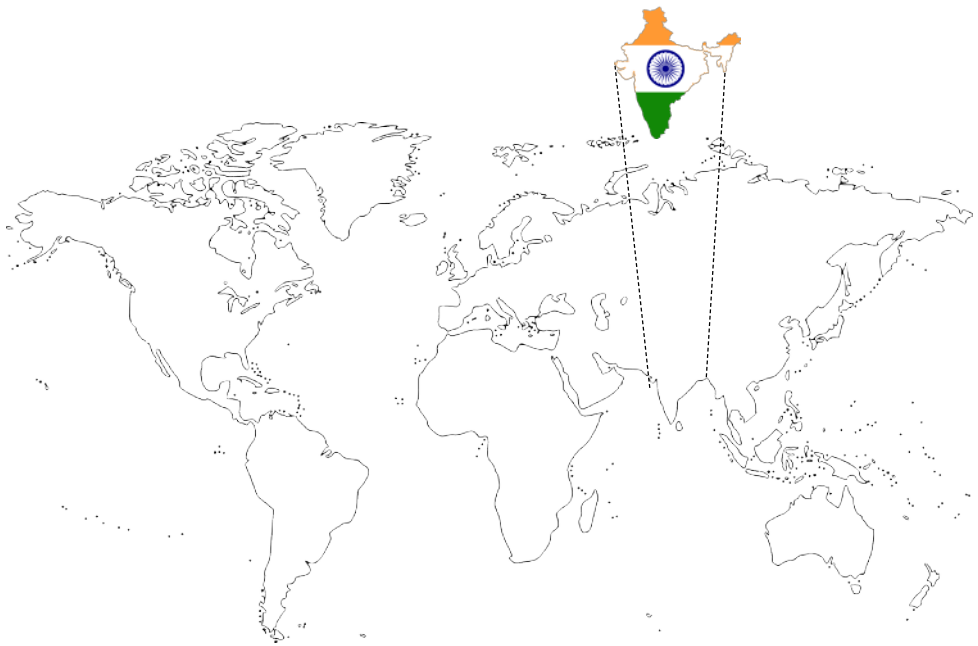








https://weatherspark.com/



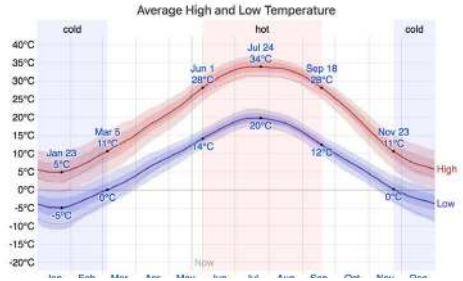


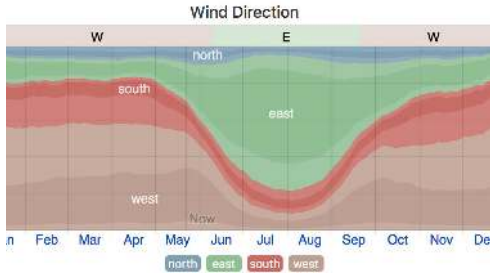


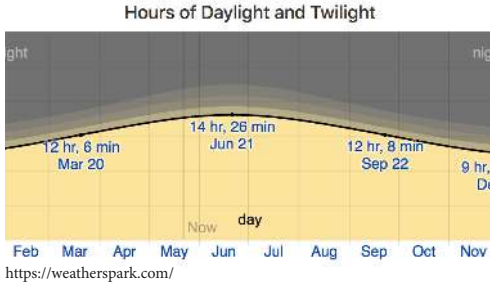


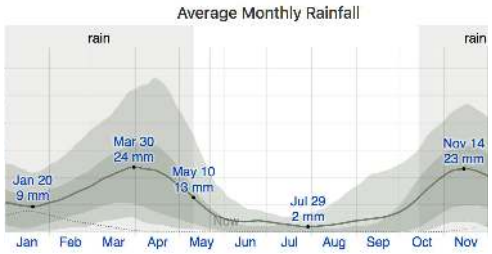


https://www.sciencedirect.com/science/article/pii/S2095263517300249





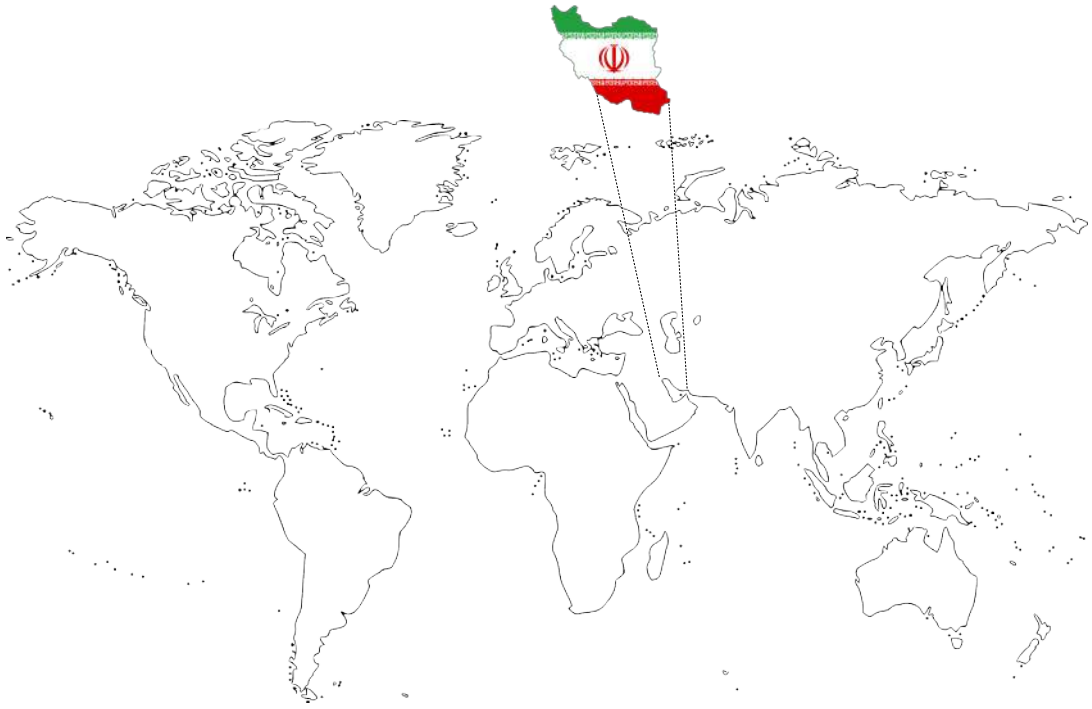




https://weatherspark.com/

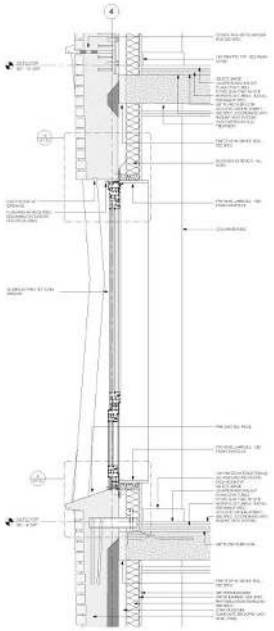
Ameri House.
Kashan ,Iran.

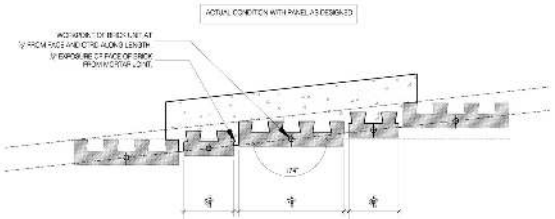
Using a central courtyard as passive cooling in the warmer months is a well know passive cooling strategy in Iran due to its high temperatures in Summer. The air is cooled in the daytime, stored in laminar layers and flows throughout the space during the day. Using a courtyard with hemp water beds allows the space to be cooled in summer. In winter when the hemp beds have no water, there is no effect and the space and can be filled when cooling is needed.



In Progress: 290 Mulberry / SHoP Architects



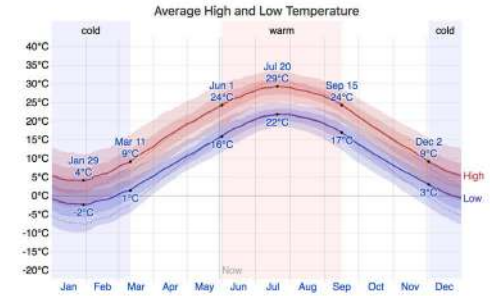


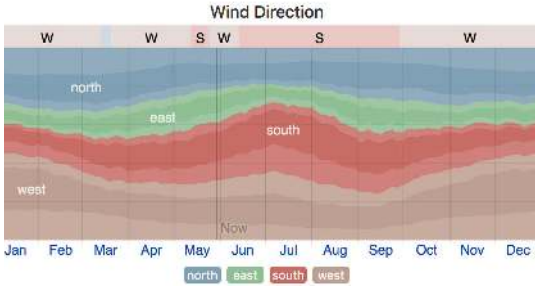


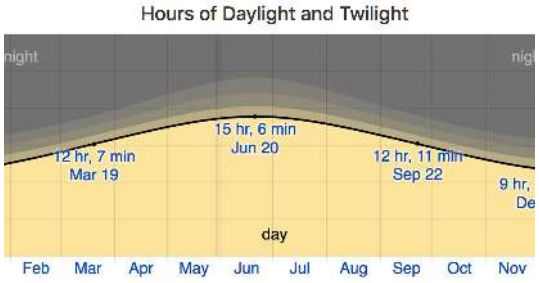
Due to strict building codes and the inability to build in the confined area SHoP architects had to create a system that could be assembled while abiding by the cities strict building codes, due to the nature of the space construction work had to be minimized.

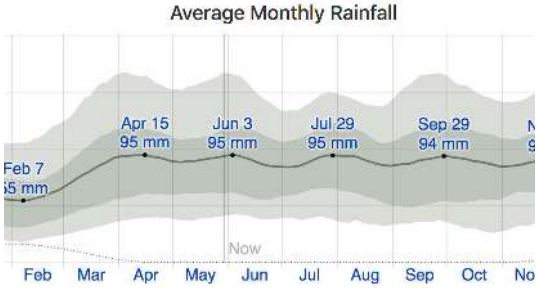
Making a literal interpretation of code written for classical ornamentation allowed us to project our enclosure over the property line at 10% intervals for every 100 square feet. Maximizing the amount of projected area, while minimizing the overall depth
of the enclosure became the key criteria for our design. When coupled with material properties and fabrication constraints, these criteria began to define an approach that was a contemporary reinterpretation of brick detailing. By customizing a standard precast brick panel system, we were able to achieve maximum effect at minimum cost. The building then becomes veiled by a textured wrapper around the street walls in contrast with the simplicity of the inner core.

https://www.archdaily.com/9028/in-progress-290-mulberry-shop-architects?ad_medium=gallery









https://weatherspark.com/









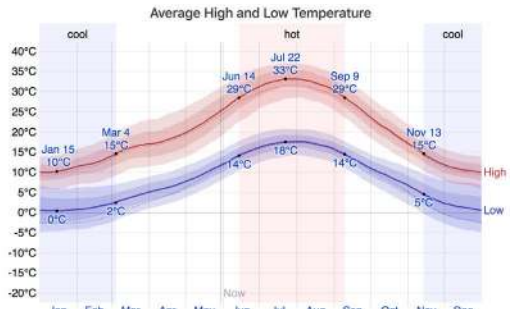


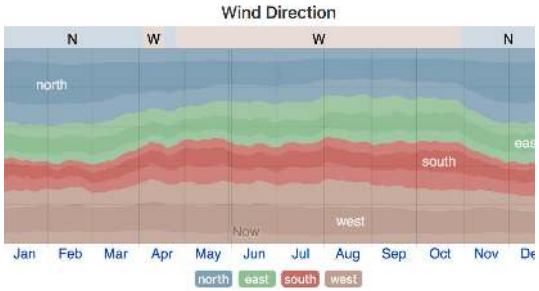
http://www.cannabric.com/

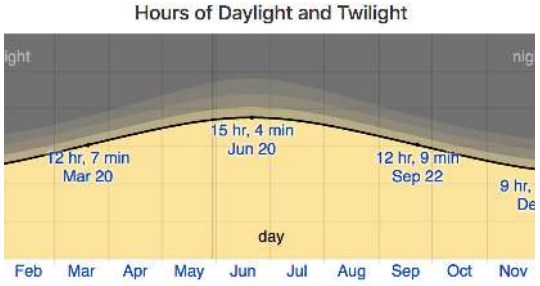
Canabric: single-family house 2005 Spain

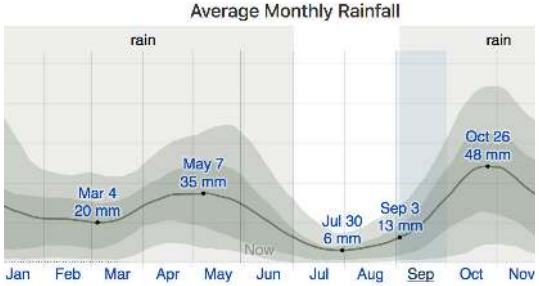
This was one of few examples were hemp bricks have been used effectively. Due to it being a private project no technical drawings are available. The project enabled us to see a real-life example of how hemp can be used as a building material. The bricks are used as a whole wall material system with only a vapour-permeable lime render placed on top to seal the bricks. Creating an organically grown building.

Spain has similar seasons to the United Kingdom allowing the performance of the material system to be of that in Spain, with added thickness to adjust to the United Kingdom's colder months.



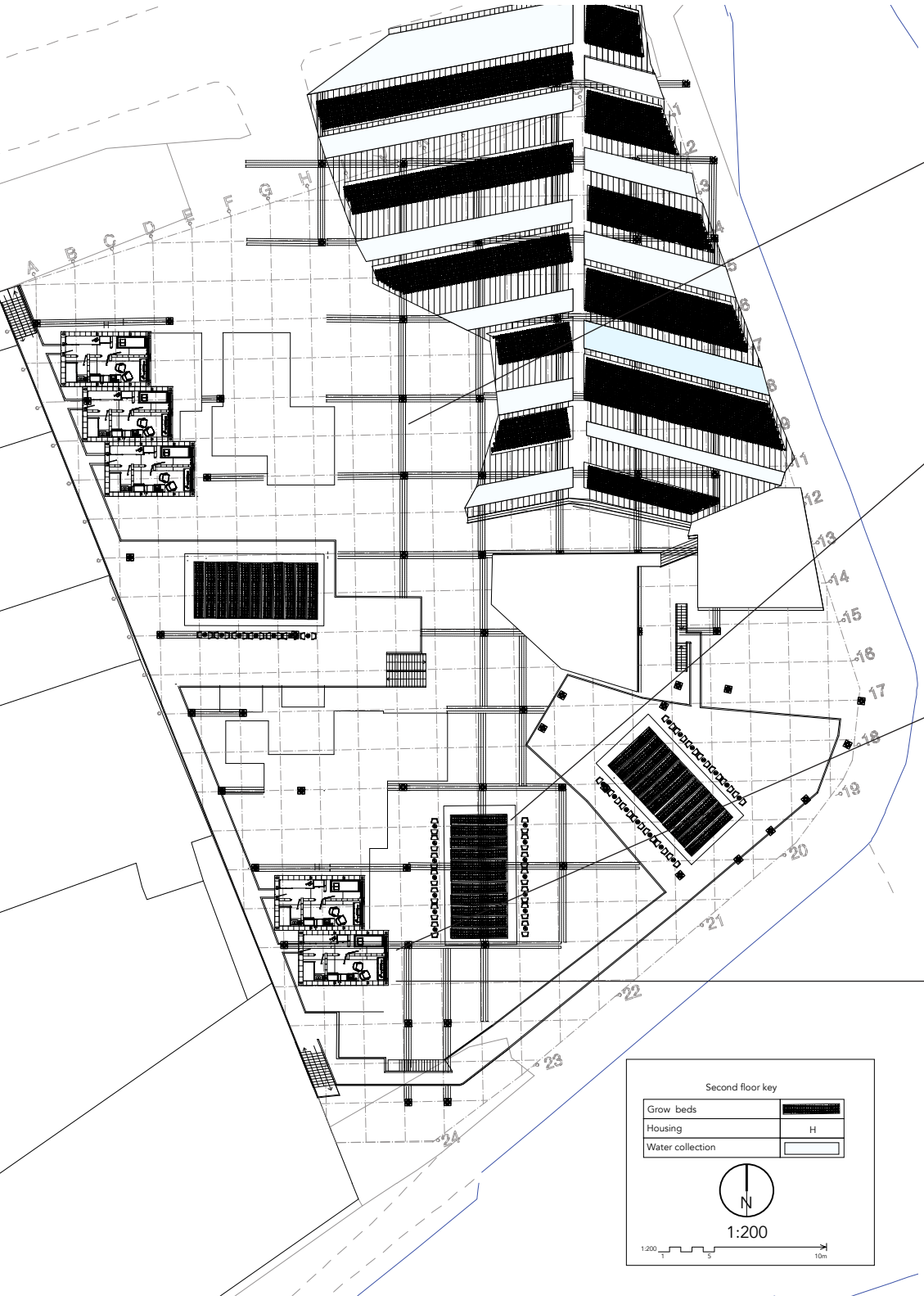






https://weatherspark.com/







India- Roof structure system



Iran- Open courtyards with water ponds



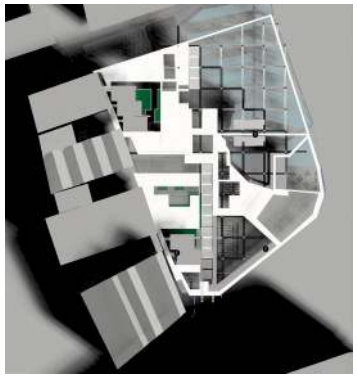
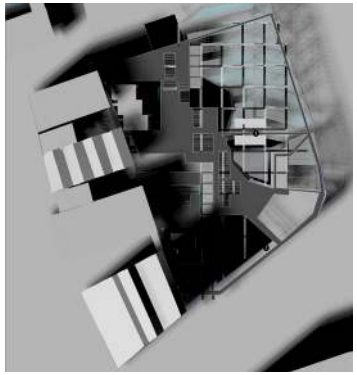
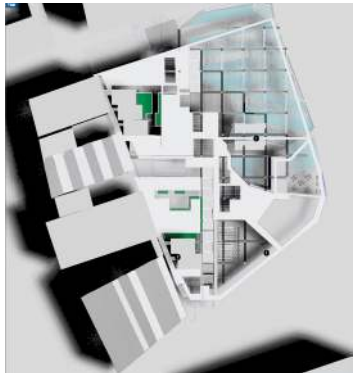
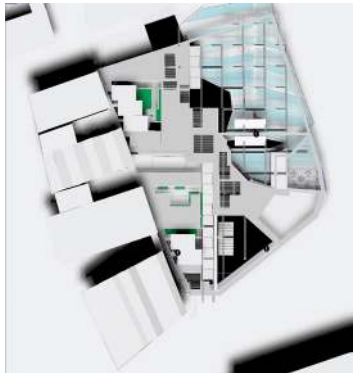
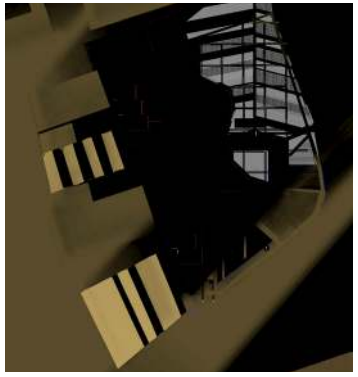
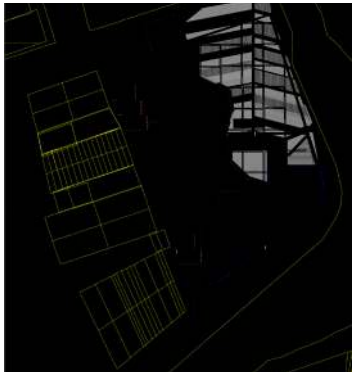
United States- Wall façade system



Spain- Canabric house

Site & Micro climate analysis

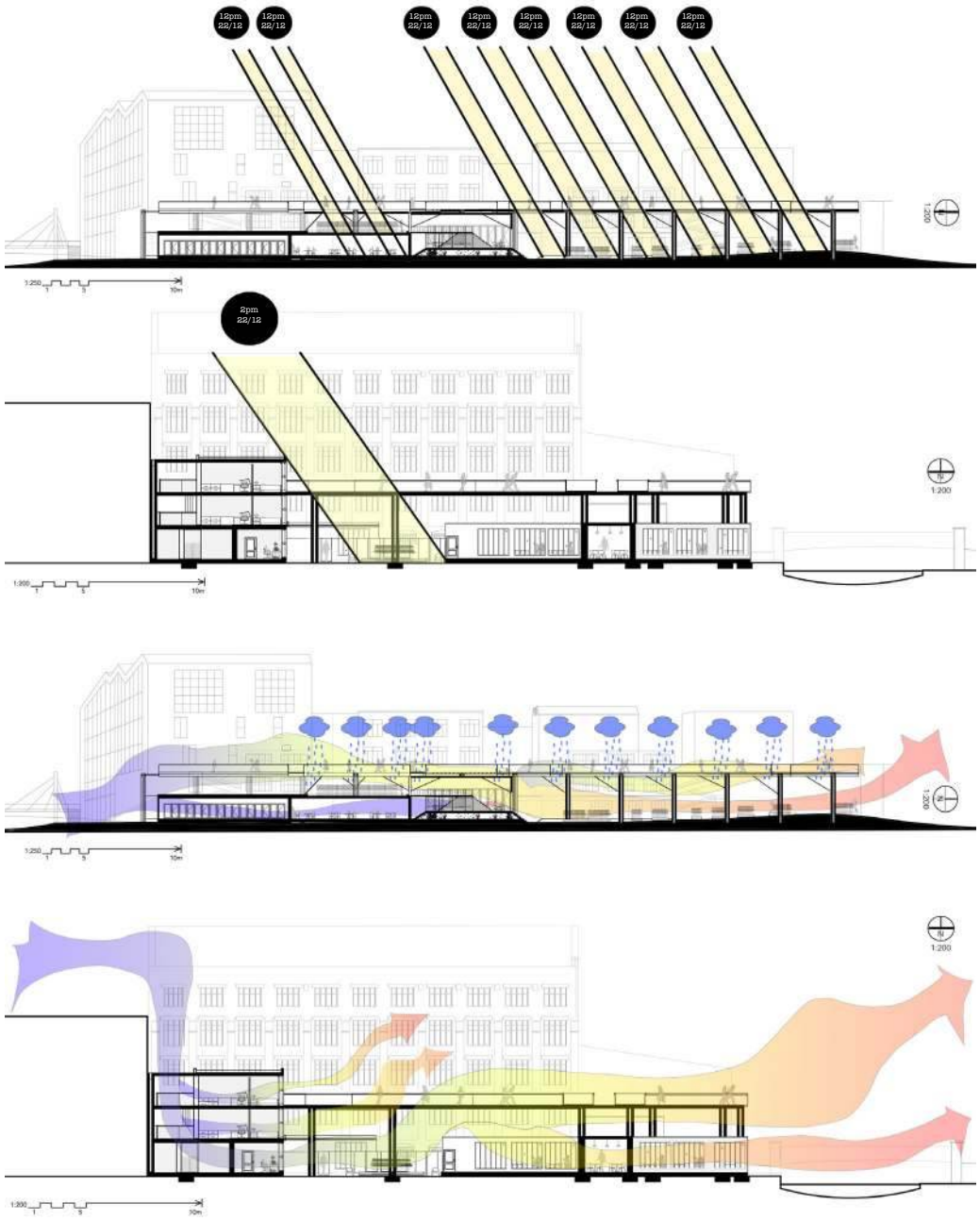
Sun path and shadows final testing

	9:00 am	12:00 pm	15:00 pm	18:00 pm
22 March Equinox solstice				
22 June Summer solstice				
22 December Winter solstice				

Light enters the site growing space at 12 pm all year for the hemp outdoor exhibition tour at the bottom of the grow bed. People sitting in the roof garden are also able to enjoy direct light at 12 pm.

The courtyard space was designed to allow light to enter at 2 pm and to allow for direct light to enter onto the grow bed, the final stage of the hemp exhibition tour.

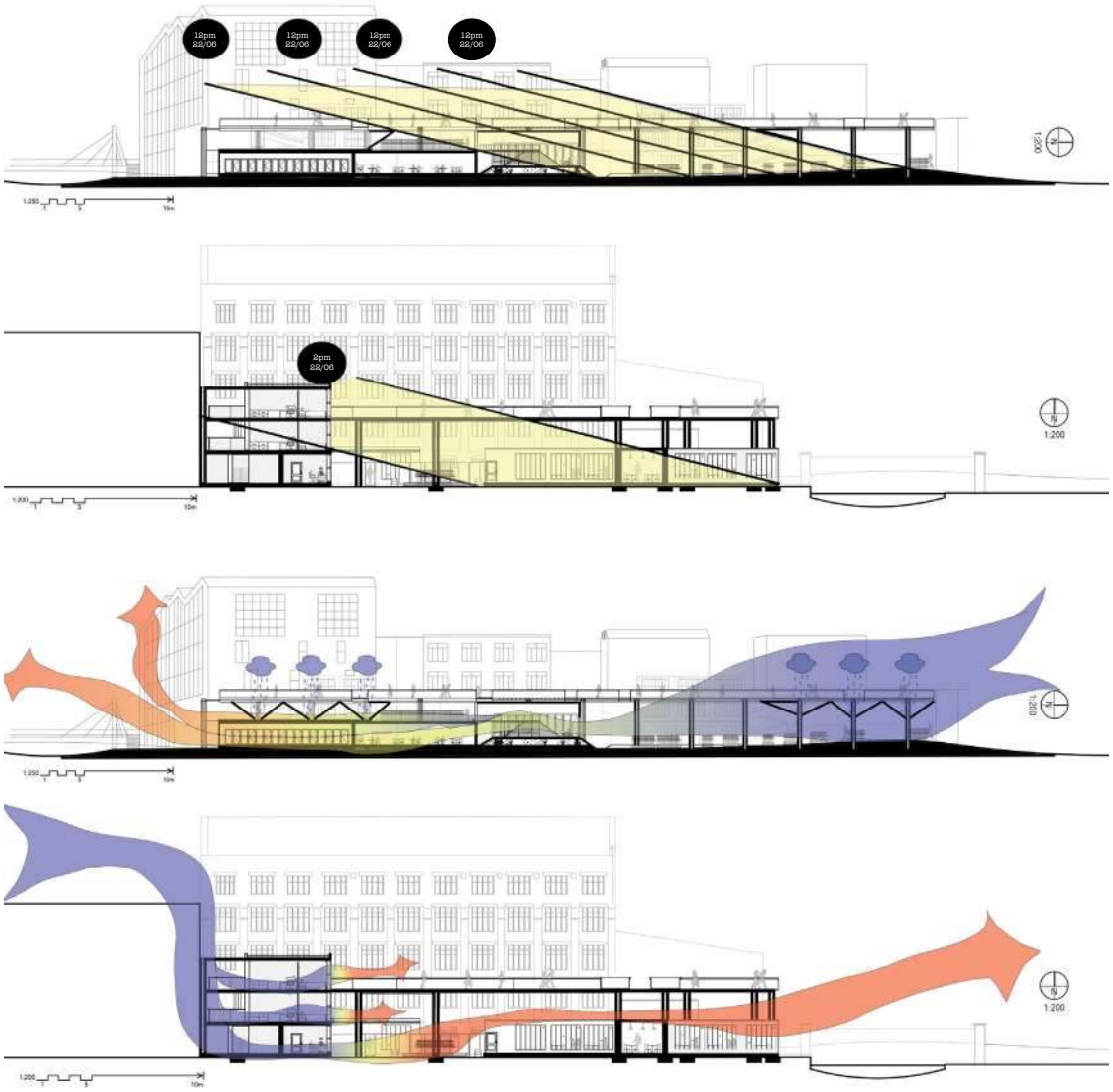
Water is collected when the structure is opened, creating shade and collecting water. The roof structure creates an atrium, and due to the wind direction coming from the left in summer, it naturally ventilates the building using stack and cross ventilation.



Low angles in the ? during the winter solstice mean the roof will be opened a lot more compared to the summer solstice months leading to more sun entering the site.

Due to the adjacent building, less direct sun enters the courtyard at 2pm but still lights up the edge of the courtyard along with sun entering other gaps between the adjacent buildings, the space will be lit at 2pm for the end of the hemp exhibition tour.

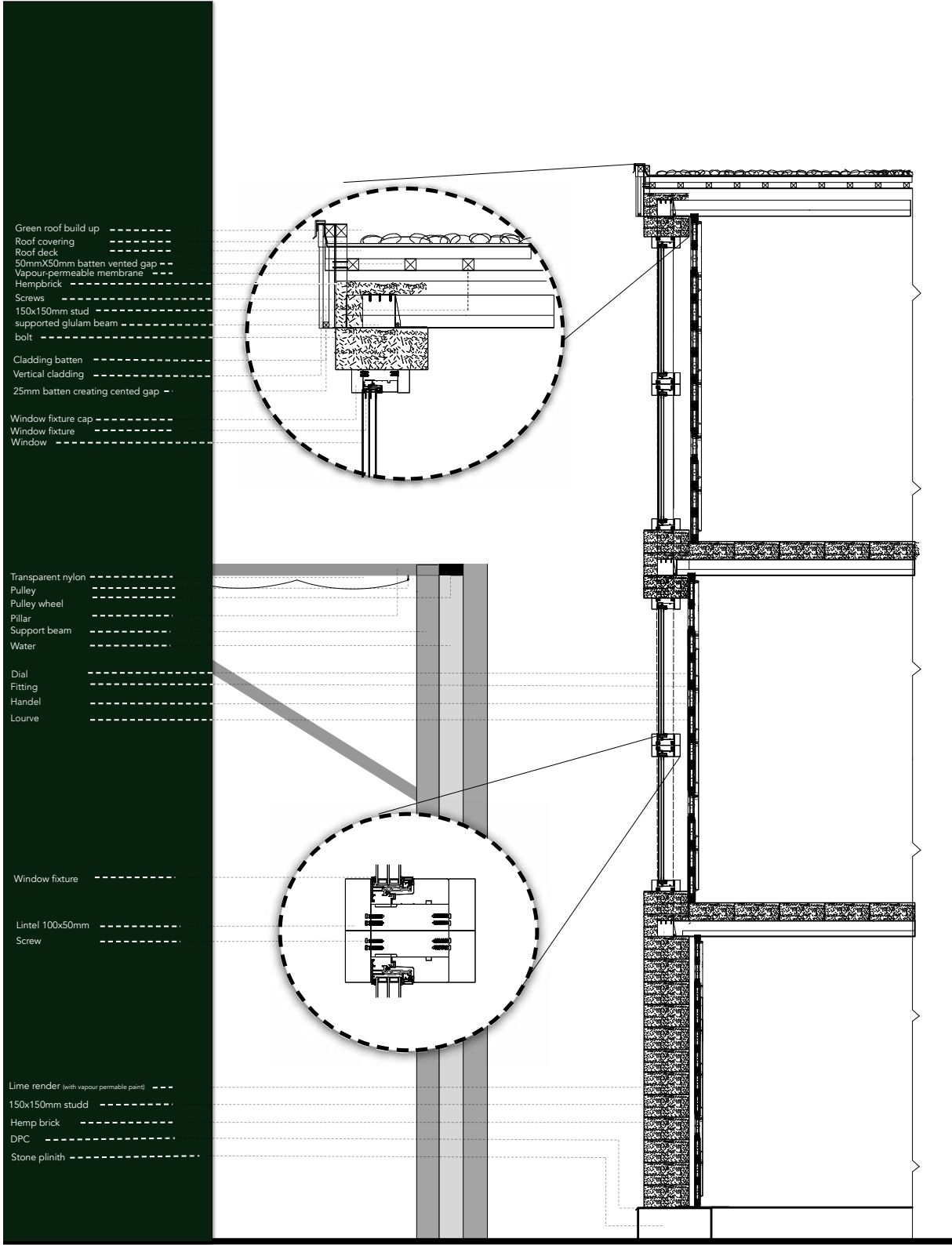
Water is collected in the winter solstice months due to high rainfall compared to summer; the winter month is crucial for rain collection and space will be adapted to ensure as much light as possible is collected.



Detail section

The material system is complemented with a louvre system to allow the space to adapt to different daily needs as the weather is unpredictable the louvre system allows space to change according to daily needs. Due to the design of the home light travels directly down the space and one window on the front elevation allows all the light in; thus, louvres are only added to the front elevation.

To prevent water damage a green roof to collect water is added on the top of the building and a stone plinth is added as the foundation to disperse any water soaking at the bottom with a damp-proof course to seal the plinth to the floor.





Hemp as a building system is made from: hemp, lime binder and water. When combined in the correct mix, they form a natural building material system. The main positives and negatives are shown on the right.

One material system to replace an entire wall system while reducing the buildings envi- ronmental impact, key to creating a natural green space that provides for both humans and planet earth. The two-main flaws are that it is not entirely waterproof and not load-bearing. The bricks can be placed between the building's structure filling in the spaces between pillars to create one whole green building. To address the waterproof problem, a closer look at a detailed section shows the entire makeup of the building system.



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Non-Load bearing



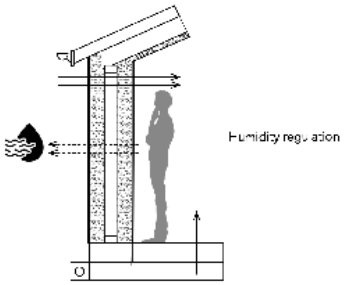
High price

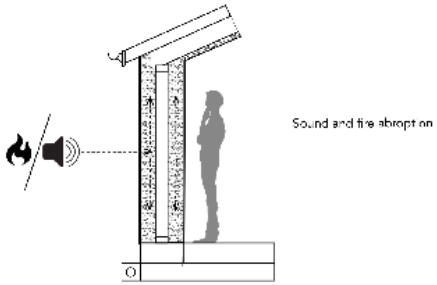


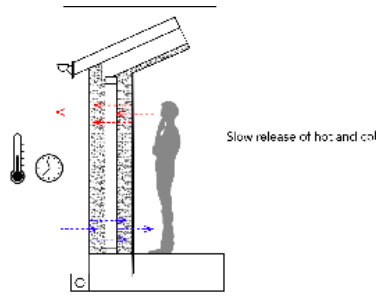
Poor water resistance



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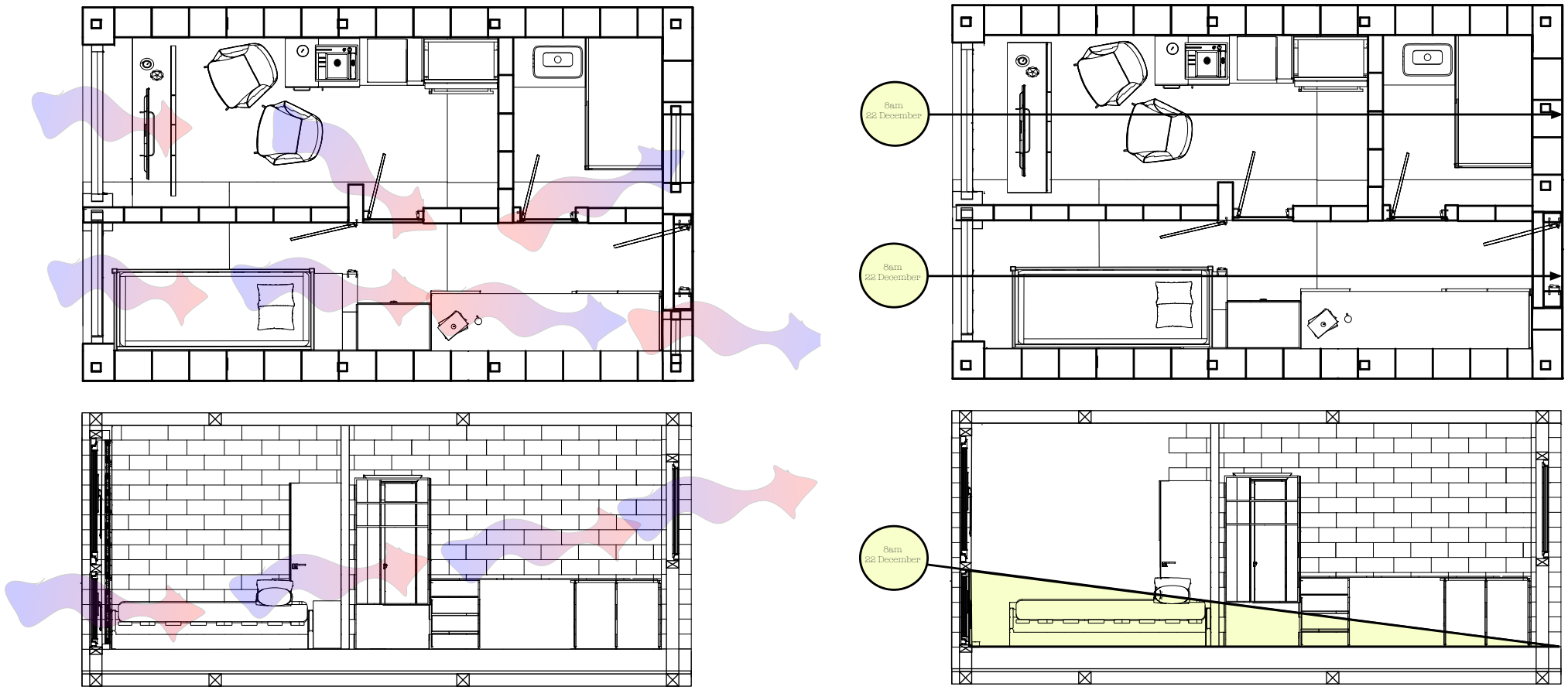


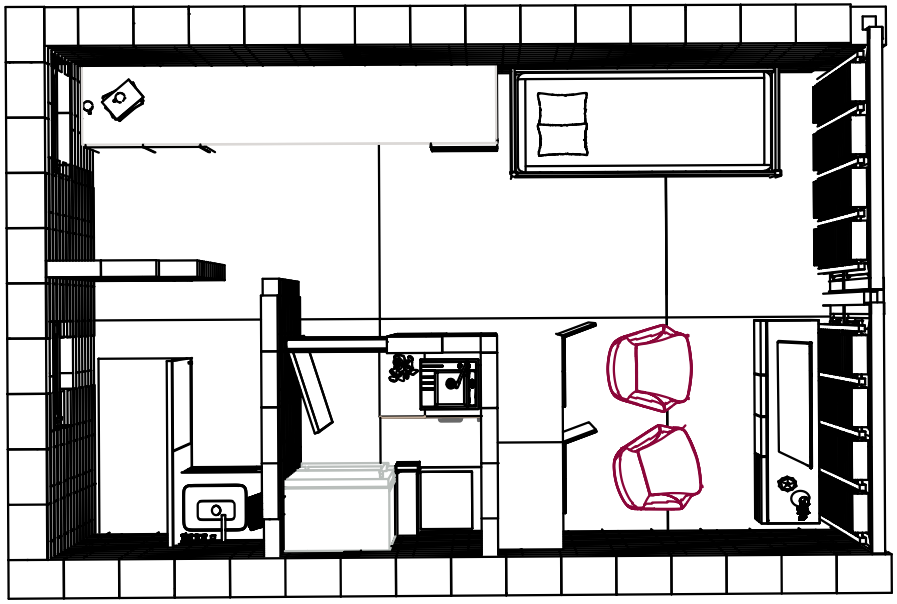


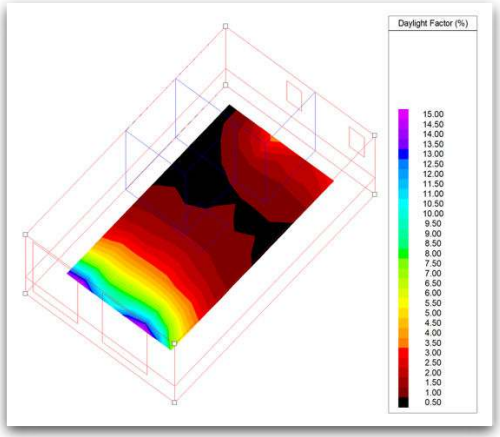


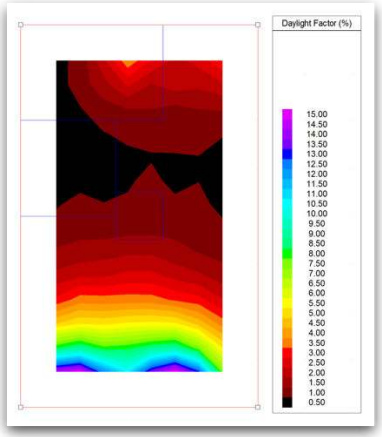
The social housing unit

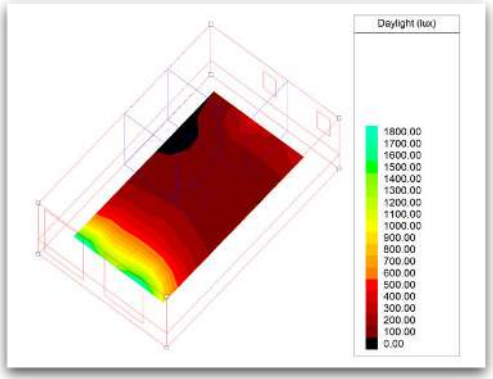
A single bedroom apartment was created for those working on site. The form was developed to allow light to travel down the space in low angle sun winter solstice months as well as allow light to enter during the entire year to wake up the individual at 8 am during summer and equinox solstice, and 9 am during the winter solstice. This was created to act as a sun alarm clock allowing natural light to wake the inhabitants.

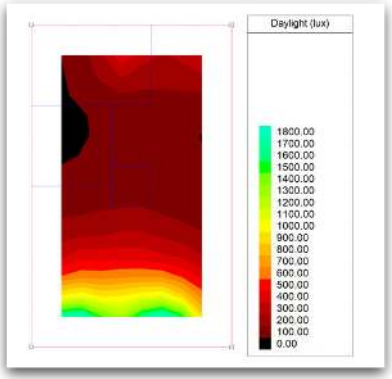


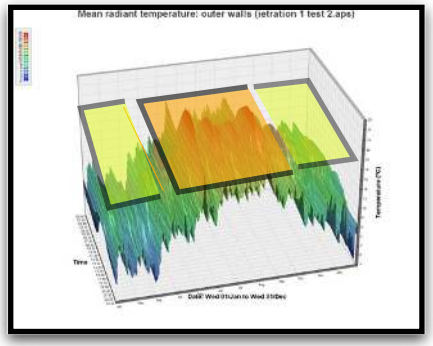


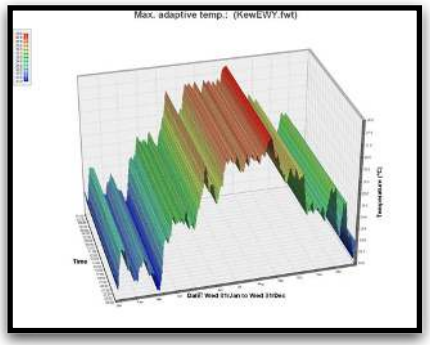


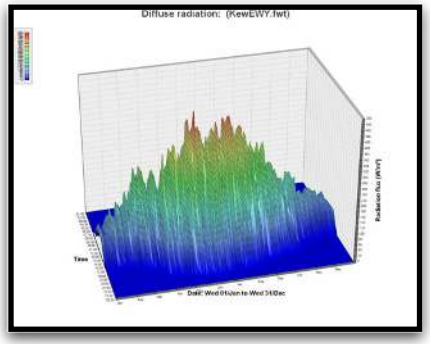


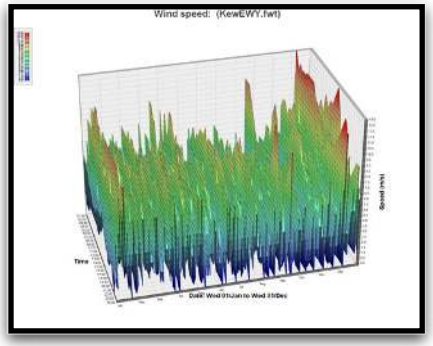








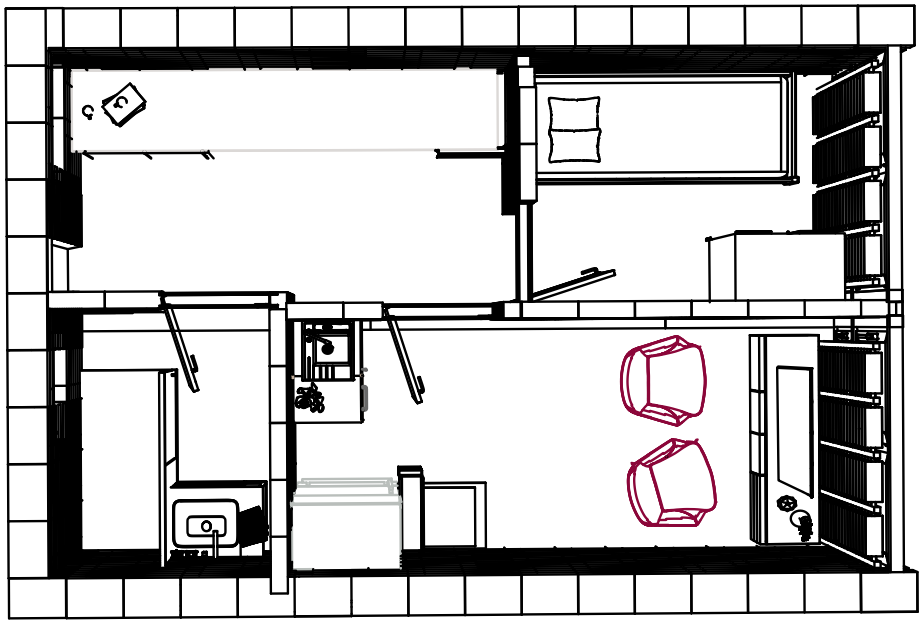


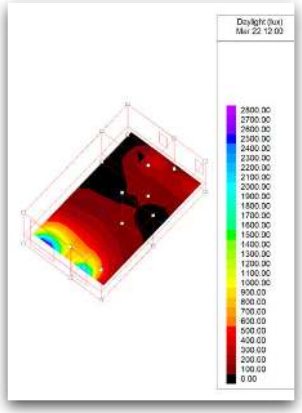


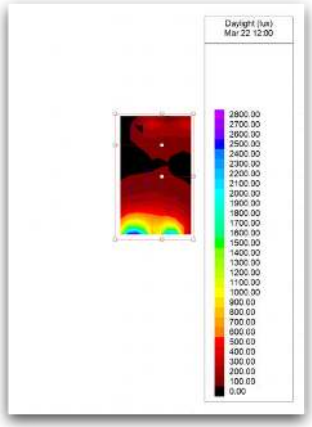
Housing unit number 1

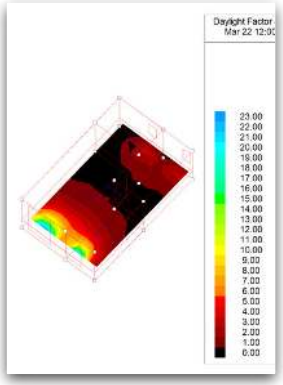
The first design uses an open floor plan to allow for a bigger space. The mean radiant temperature showed an interesting out-come and the open space is not as good. Higher temperatures in summer and lower temper- atures in winter, this result is not desirable but shows clear room for im-provement.

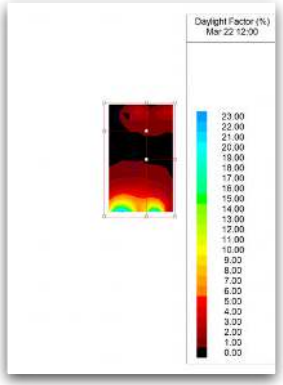
The high daylight factor and LUX level in-dicates the room is well lit but can be seen as too bright in areas. Light travels clearly through the entire space

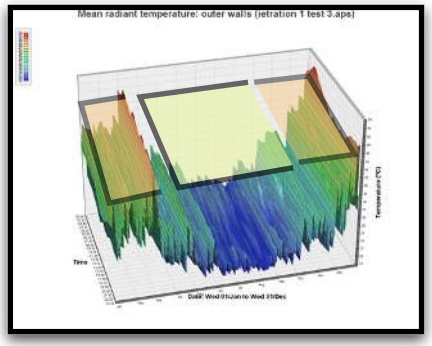


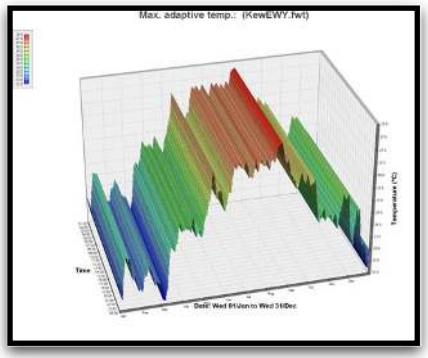


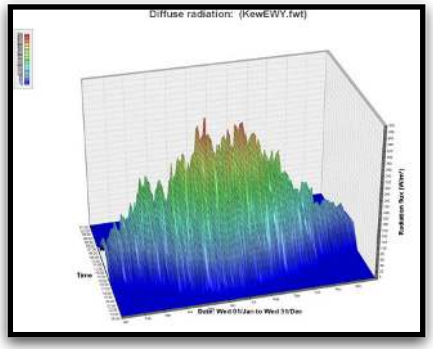


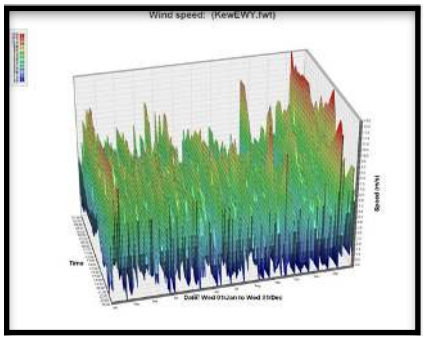








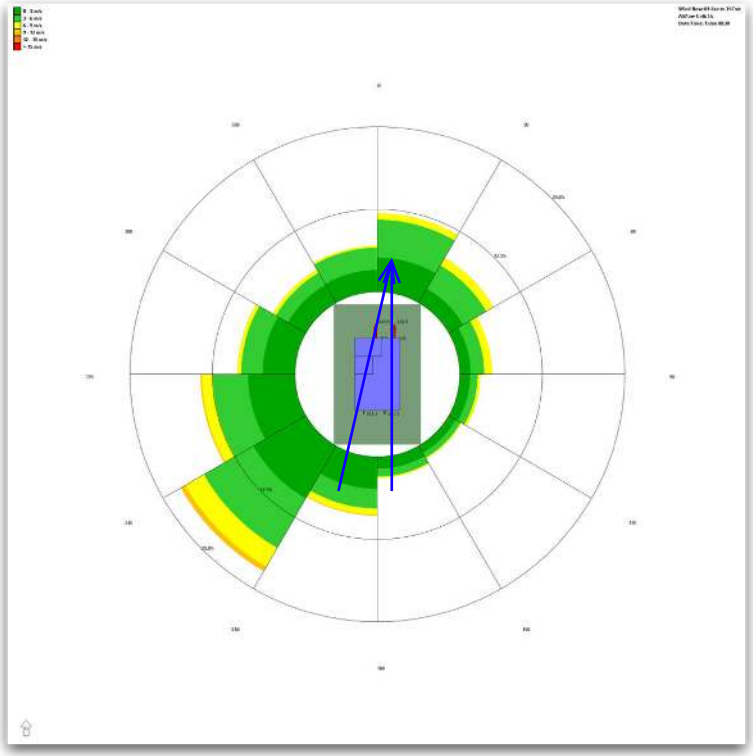


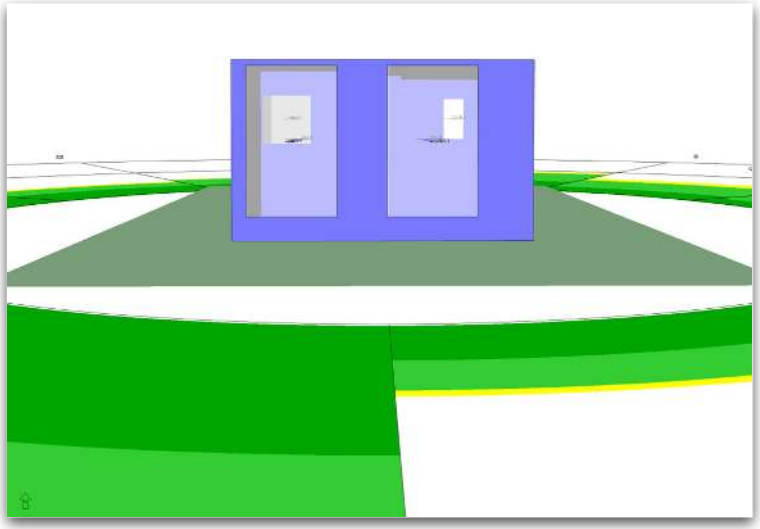


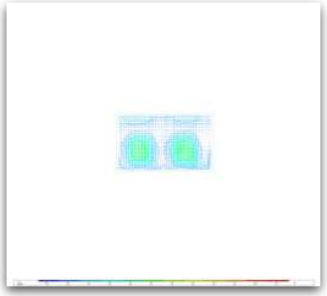
Housing unit number 2

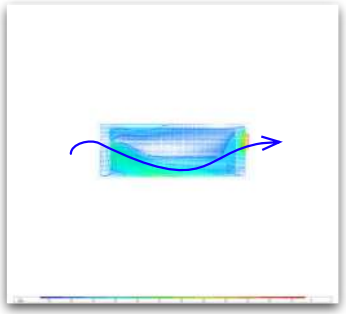
The second design uses a closed floor plan to allow for a confined space. This space had different results with a high mean radiant temperature in winter and low mean radiant temperature in summer which is preferable in the United Kingdom. Showing that hempcrete performs better within confined spaces regu- lating internal conditions effectively due to its hygroscopic nature.

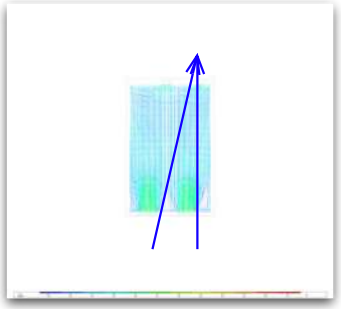
The daylight factor and LUX level indicates the room is well lit but has issues in areas that aren't close to the windows. Light travels through the space but doesn't perform as well as number 1.

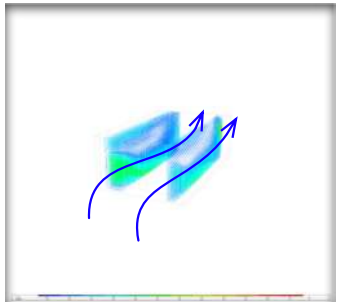






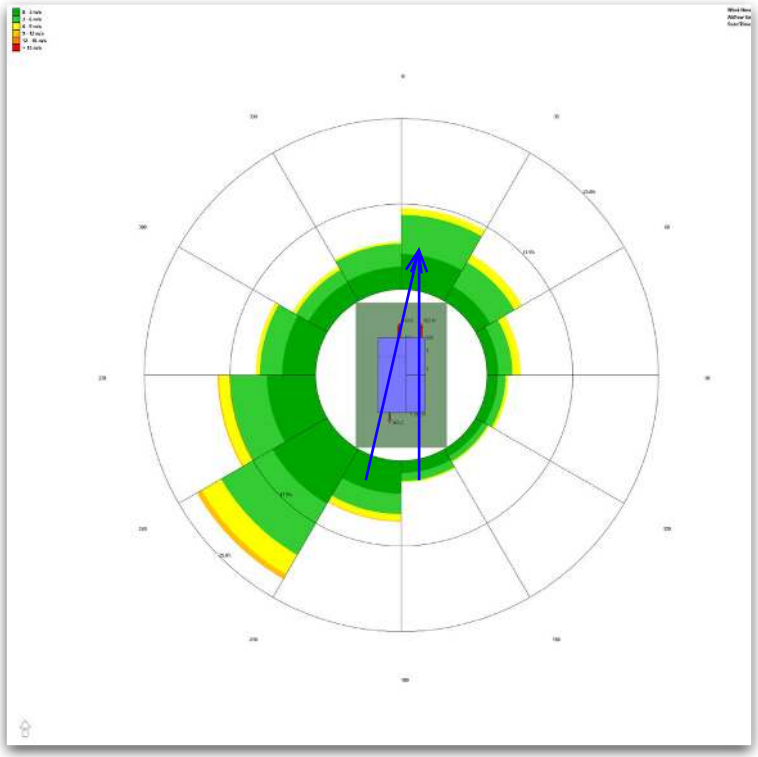


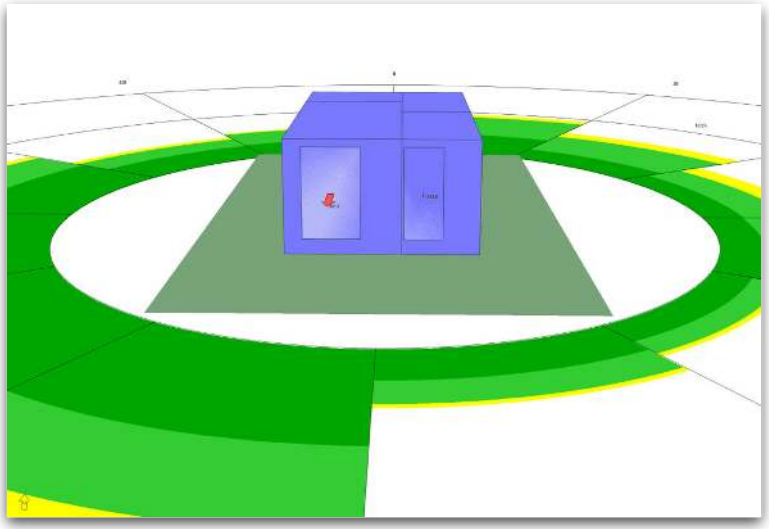


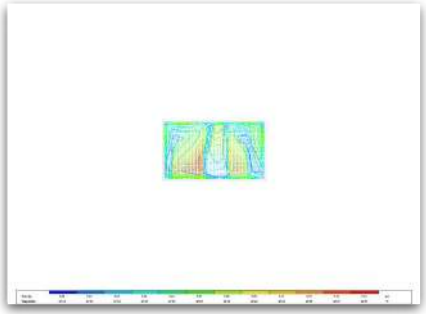


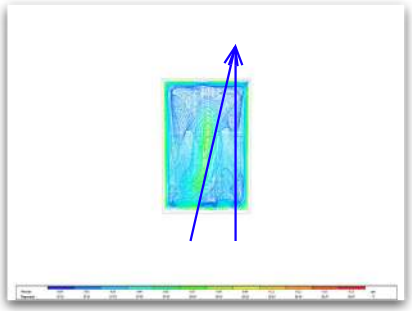
Housing unit number 1

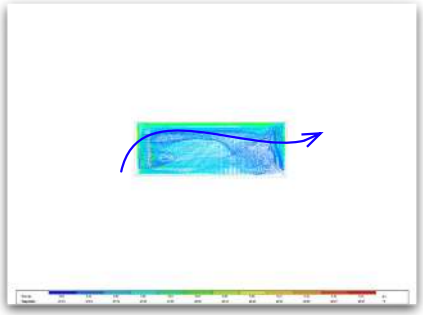
Adding windows on either side allows for cross ventilation. The open space had lower wind levels with the wind being dispersed due to the open space nature. The wind travels from the floor and goes up; this may cause issues when walking around or working due to the wind currents coming from the floor.

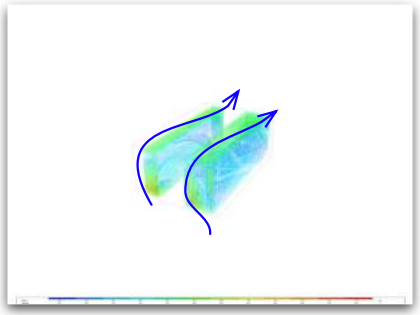












Housing unit number 2

Two opening cross-ventilation ?? lets air pass through the space freely. The closed space allows air to travel above the inhabitants and has less effect on those using the space, airing the unit and leaving with no issues from wind currents.

This concluded the choice for the unit. Unit number two was chosen due to: sufficient light levels, mean radiant temperature performance and wind currents travelling along the roof.

Conclusion

The assessment of the effectiveness of hempcrete concludes to find that hemp can be used as a material system. Using a closed wall plan, the bricks lead to airing of the room due to its hydroscopic nature, allowing clean air to travel through the space maintaining desired temperatures in summer and winter, while allowing cross double sided ventilation to transfer air and space formed to fit the sun.

A closer look into sections shows the make-up of a hemp brick material system with a perspective view plan to show each space, with Unit 2 being developed into the final outcome due to the results explained.

Strategies used to combat the problems highlighted in the analysis through form and function. High rainfall can be combatted with collection systems, solar and thermal issues can be tackled using passive strategies, a polluted city can be combatted with green building,? whole in nature built by nature.

In conclusion, a passive and active solar-thermal natural design building can be built and allow for an off-grid building to be built in the United Kingdom. Using natural elements to bring various benfits to people and planet earth.

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This report concludes that hempcrete can perform effectively as a material system for Back to Basics, allowing a closed hempcrete wall to regulate air and temperature passively across the seasons. Combined with passive solar, rainwater and cross-ventilation strategies, the findings support a fully off-grid, passive building strategy for Hackney Wick.



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